AthenaK shock encounters Implementation and test report

Recorded local evidence | 8 October 2026 | Branch: codex/shock-multiphase

What was implemented

On-the-fly shock detection and persistent per-particle shock-encounter histories. The live C++/Kokkos detector samples the completed fluid state; the tracker groups particle observations into episodes; migration, output and restart preserve those episodes.

Component	Purpose
Live mesh detector	Port the frozen ShockIt numerical contract and retain the unreduced shock region.
Encounter tracker	Record entry/exit brackets, positions, Mach estimates, normals and validity/censoring flags.
Persistence	Move complete typed state with each particle; use bounded event output and source-bound restart sidecars.

Current result

Local engineering checks pass. The simple Mach 2 pilot records 128 completed episodes for 128 particles in each of eight seeds. Strong and boosted fixtures expose additional episodes and strength bias. Human scientific approval and R1 release acceptance remain pending.

How to read this report

Pages 2-8 explain the test setup, methods, measurements, persistence, performance and reproduction. The following 28 figure pages retain the original vector plots and add plain-language reading notes and case-specific results.

The existing lagrangian_mc tracer makes stochastic parent-cell hops. Positions are cell centers. Lines in trajectory plots are sample sequences, and timing brackets describe neighboring shock-region observations. Neither is an exact continuous material crossing.

Evidence source: production commit 3de18d97; selected base multiphase_turb 526a187f938e38634e89a88192ebe9731c4d36bb. This report adds explanation to recorded results; no simulations or numerical thresholds were changed to produce it.

2. The primary physical experiment

A translating planar shock is evolved in a thin two-dimensional domain. Independent Rankine-Hugoniot jump relations define its initial states and known speed. This is genuine 2D fluid output, but the planar setup does not qualify curved or intersecting shocks.

Setting	Pilot configuration
Domain and grid	x: 0 to 1; y: 0 to 0.1; 128 x 4 cells; no AMR.
Physics and numerics	Newtonian ideal-gas hydro, gamma=1.4; RK2; PLM; HLLC; CFL=0.3.
Shock	Mach 2; initial front x=0.3; speed=0.5; x_front(t)=0.3+0.5t.
Particles and duration	128 tracers seeded in x=0.45 to 0.75; t_limit=0.25.
Boundaries	Outflow along x; periodic along y.
Diagnostic settings	width=4 cells; min_mach=1.1; capacity=4096; trace=true. These are pilot settings, not approved scientific tolerances.

Independent conservation reference

For upstream Mach M and gamma g, the density ratio is (g+1) M^2 / ((g-1) M^2+2), and the pressure ratio is (2g M^2-(g-1))/(g+1). The fixture checks mass, momentum and energy conservation in the shock frame before running the simulation.

At M=2, g=1.4: density ratio=2.6666667 and pressure ratio=4.5. Upstream density and pressure are both 1. The downstream values are 2.6666667 and 4.5. Velocities are -1.8664319 upstream and -0.3874120 downstream.

The pressure and density plots therefore have a reference independent of the detector. P/rho is a temperature proxy in code units. The Eulerian density-midpoint location is a front diagnostic; it is not used to invent particle crossing times.

Why particles encounter this front

The front moves right while the upstream fluid and sampled tracers move predominantly left. Their histories meet the resolved shock region. Staying inside that region for many timesteps should create one numerical episode, not one event per timestep.

3. Engineering tests: what each checks

The recent integration suite has 20 subprocess runs. Some must succeed; others must abort with a specific diagnostic. All produced the expected outcome. Test IDs refer to the implementation plan and recorded checks.

Test / IDs	Method and recorded result
Non-interference CFG01, PHY08, SYS02	Run identical uniform and planar problems with tracking on/off. Particle observations are exact matches; 17-digit conserved slices are byte-identical.
Sampling INT06, INT08	Compare final-step shock samples with native particle thermo history. Generation, time, position and density agree exactly.
Negative controls CFG06, PHY09	Uniform flow, contact at constant pressure/velocity, and no particles. All emit zero events.
Local migration INT01, IO01	Use multiple local mesh blocks. All initial IDs survive; 113 particles change parent block. This is not an inter-rank runtime test.
Birth/censoring EVT02, EVT05	Seed initially or later inside the shock region. Missing valid pre-entry evidence is marked left-censored.
All-phase restart RST01, RST07	Checkpoint outside, active and pending-exit phases. Resume and compare exact particle rows and canonical events.
Damage/replay RST03/04, IO08	Remove/corrupt sidecars and damage retained history. Expected explicit rejection; immutable replay checks prevent conflicting event timelines.
Capacity / configuration IO03/04, CFG	Force capacity=1 and an overflow; request unsupported RK3. Both explicitly reject instead of silently dropping state or accepting unqualified integration.

The native encounter unit also passes state-transition checks. Separate earlier evidence contains 12 explicit-rejection runs, 4 replay runs, 18 actual-method boundary cases (with some simplified harness context), and 13 actual production domain-exit records. These categories must not be summed as independent physical shock passages.

4. Detector parity and independent transport

ShockIt parity: fidelity of the numerical port

53 fixtures compare the native detector against the frozen ShockIt local_pass reference. Shock-region masks must match exactly. Floating quantities use unchanged absolute and relative tolerances of 1e-12. All 53 pass.

The fixtures exercise constant fields, ideal shock jumps, gradients/normals, estimator thresholds and invalid inputs. Matching the reference validates the port; it does not independently certify that ShockIt or the encounter association is scientifically correct.

Uniform transport: an independent physical probability model

For resolved positive uniform velocity u, the expected cell-hop probability per step is p=u*dt/dx. Summing the Bernoulli hops gives independent reference mean sum(p) and variance sum(p*(1-p)). This uses physical flux probabilities, not the detector or RNG implementation.

Quantity	Reference	Measured
Particles / steps		1024 tracers / 117 steps
Mean cell hops	16.000000	15.896484
Hop variance	13.806057	12.331405
Mean / variance deviation	Existing +/-6 standard-error diagnostic	-0.89 / -2.40 standard errors

Both diagnostics pass. TRC11 also checks 119,808 actual transfer records: source parent, step start/end times, probabilities, random draw, selected direction and realized hop. Maximum probability discrepancy is 6.11e-16 against the unchanged 1e-12 tolerance.

This fixed-seed ensemble check is not proof of RNG independence at research scale. The inherited additive RNG key has alias/correlation questions that remain open. There is no deterministic per-particle material-path oracle.

5. Physical recovery and strength limitations

All 21 held-out fixtures executed. They vary eight Mach 2 seeds, Mach=1.2/1.5/3/5, gamma=5/3, stationary/boosted/reflected fronts, nx=64/256, CFL=0.15/0.45 and one thin-3D case. Run success is distinct from physical acceptance.

Fixture	Complete	Unique IDs	Suspended
Mach 2: each of eight seeds	128	128	0
Mach 1.2	128	128	0
Mach 1.5	127	127	1
Mach 3	128	128	0
Mach 5	134	128	0
Boosted Mach 2	132	127	1
Other gamma/speed/direction/resolution/CFL/thin-3D cases	128 each	128 each	0

Mach 5 gives six additional episodes for IDs 12, 43, 47, 57, 74 and 115. Boosted Mach 2 gives five additional episodes for IDs 3, 52, 67, 80 and 102, plus one suspended encounter. These are preserved limitations against the simple single-front expectation; stochastic returns and weak region lobes need scientific interpretation.

The boosted case adds 2 to the fluid and front velocities, retaining Mach 2. Its differing counts and bias expose frame dependence that needs qualification.

Case	Pressure-Mach error range
Mach 2, seed 11	-0.0163 to +0.0118
Mach 1.2	-0.1031 to -0.0157
Mach 5, all episodes	-3.9657 to +0.0594
Boosted Mach 2	-0.1091 to -0.0771

Error means estimated Mach minus prescribed Mach. Strength uses the first valid inside sample, with fixed endpoints that are not certified plateaus. Weak secondary episodes remain in the Mach 5 range. No outliers or tolerances were adjusted for acceptance.

6. Encounter timing, output and restart

An episode begins from valid unreduced-region membership, with an upstream-state entry guard where prior evidence exists. A first outside observation starts pending exit; two consecutive valid outside samples close/rearm. Earlier re-entry merges and flags flicker. Invalid samples cannot close an episode.

For valid pre/post evidence, entry bracket=[pre.time, first.time] and exit bracket=[last.time, post.time]. These bound adjacent sampled membership changes, not an exact material shock crossing. Cadence limitations, uncertain endpoints and direction ambiguity are explicit flags. Birth inside is left-censored; an unfinished run leaves a separately readable suspended right-censored view.

Restart phase	Generation	Exact resumed rows	Exact events
outside	1	41,600	128
active	61	33,920	128
pending	68	33,024	128

The strict reader detects no duplicate canonical event identities in these runs. Event keys are (lineage, unsigned 64-bit particle ID, unsigned 64-bit sequence). Complete encounter state migrates/compacts with the particle. Actual outflow records are emitted before particle deletion.

Per-generation event chunks have bounded capacity and immutable commits. Restart pairs the native checkpoint with a full encounter sidecar and requires the retained history prefix. Checks bind implementation, configuration, time, count, native layout and partition; missing, changed or incompatible data reject explicitly.

This is same-source, same-layout restart evidence. It is not an all-rank atomic transaction or power-loss/storage-fault qualification. MPI builds/singleton checks pass; the local two-rank launcher failed before AthenaK because of runtime compatibility. True distributed migration/restart and target filesystems remain untested.

7. Measured laptop performance

Uniform flow, 1,024 tracers, t_limit=0.1; two meshes and three modes. Three repetitions per mode/mesh, with execution order rotated. All 18 benchmark runs completed. Times below are whole-process medians, including startup and ordinary outputs.

Mesh	Tracking off	Tracking on	Tracking + trace
64 x 4	0.01163 s	0.04171 s (3.59x)	0.28880 s (24.84x)
256 x 4	0.04174 s	0.12862 s (3.08x)	1.10772 s (26.54x)

Tracking costs about 3.1-3.6x in these tiny jobs. Full observation/transfer traces cost about 25-27x. Durable empty event chunks are still committed every generation; trace mode writes every sampled observation and transfer. These costs must be visible when choosing operational output settings.

Output-cost figures report actual shock diagnostic bytes and file counts, excluding fluid/thermo output. Resident memory was not measured. Short local jobs do not establish sustained production overhead, scaling, MPI/GPU performance or costs on RUSTY/Frontier.

Reproduction evidence and report scope

The initial fresh engineering and physical suites took 13.83 and 28.05 seconds. Their raw evidence occupied about 1.2 GB. The plotted workflow was independently repeated without reuse in a fresh directory: all 21 physical measurement sets and 27 PNGs reproduced exactly, including output cost. The runtime figure retained fresh measurements.

The first plot attempt hit a Matplotlib layout error on an all-zero restart axis. Only the display scale/label coordinates were fixed; exact-equality assertions were retained. A packaging count was corrected from 26 to 27 matching figures because the output-cost plot also reproduced. These were reporting issues, not numerical fixes.

Remaining gates

Human review and acceptable recovery/duplicate/timing/strength thresholds; RNG and frame dependence; sustained encounters and turbulence; true inter-rank operation; threaded/GPU/single precision and cluster budgets. AMR, cooling/source terms, MHD and relativistic physics require separate qualification. Material loops/strings are excluded.

8. Run the tests and locate the evidence

Run from the development worktree using the existing Release executable. Use new work and output directories each time. The existing ShockIt Python environment contains NumPy, Matplotlib and h5py; the PDF authoring step uses the separate bundled document runtime, with ReportLab and pypdf.

```
cd /Users/hitesh/.codex/worktrees/shock-multiphase/athenak_nyu
MPLCONFIGDIR=/tmp/athenak-mpl \
/Users/hitesh/hitesh/git/shockit/.venv/bin/python \
  tst/shocks-mc/laptop_visuals.py \
  --executable build-release/src/athena \
  --workdir /tmp/athenak-report-run-new \
  --output /tmp/athenak-report-plots-new
open /tmp/athenak-report-plots-new/index.html
```

The existing plot runner generates the raw tests, gallery and plot-only PDF. To produce this explained report from a new packet, run the following separate step:

```
/Users/hitesh/.cache/codex-runtimes/\
codex-primary-runtime/dependencies/python/bin/python3 \
  tst/shocks-mc/explained_pdf.py \
  --packet /tmp/athenak-report-plots-new \
  --plots /tmp/athenak-report-plots-new/all-plots.pdf \
  --output /tmp/athenak-report-explained.pdf
```

That PDF step consumes existing results; it does not rerun the simulation. For a fresh build: cmake -S . -B build-release -DCMAKE_BUILD_TYPE=Release -DAthena_SHOCK_TESTS=ON; then cmake --build build-release -j8.

Source and evidence identity

Core implementation fingerprint:

```
3feb3bcdf77d21296516544e67b5ad75f6ac6ca7bc17c4b48be886ddc0fb7cbc
```

Executable SHA256:

```
0b78478d9b550584051c8a564c562d9bf76628d88f66ddb3ab24aaaa368aff2d
```

Pilot input: inputs/tests/shock_mc.athinput

Test runners: tst/shocks-mc/integration.py, qualification.py, laptop_visuals.py

Recorded packet: evidence/shocks-mc/laptop-visuals/

Parity/persistence review: evidence/shocks-mc/review3/

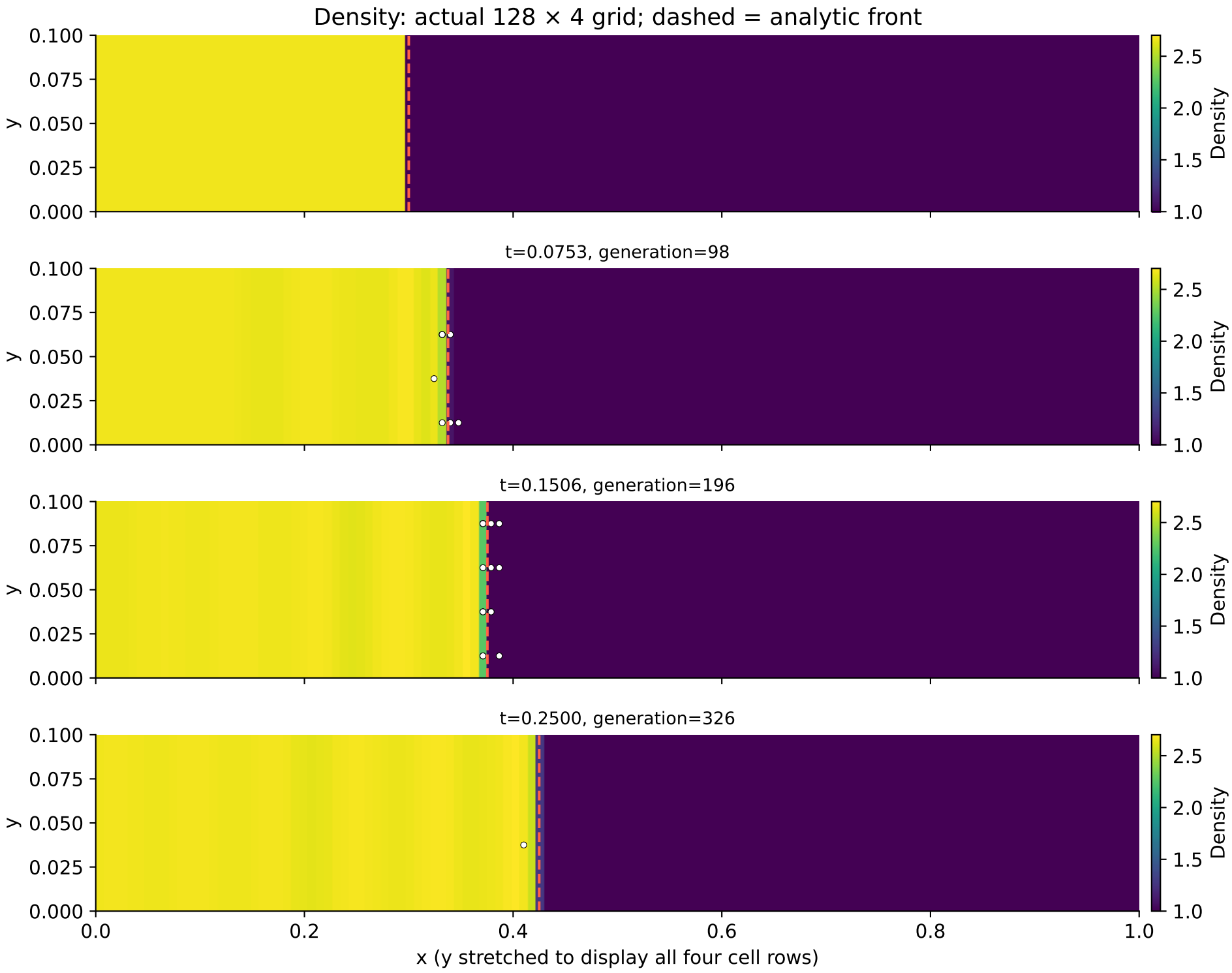
Schema/support: docs/shocks-mc/schema.md, README.md, human-review.md

Recorded versions: Python 3.14.0, NumPy 2.5.0, Matplotlib 3.11.0, h5py 3.16.0

Scientific sign-off: pending. CPU Serial/double/RK2/uniform Cartesian ideal-gas hydro is the measured local scope. Earlier evidence retains its own source identity.

Figure 01 | 2d dens

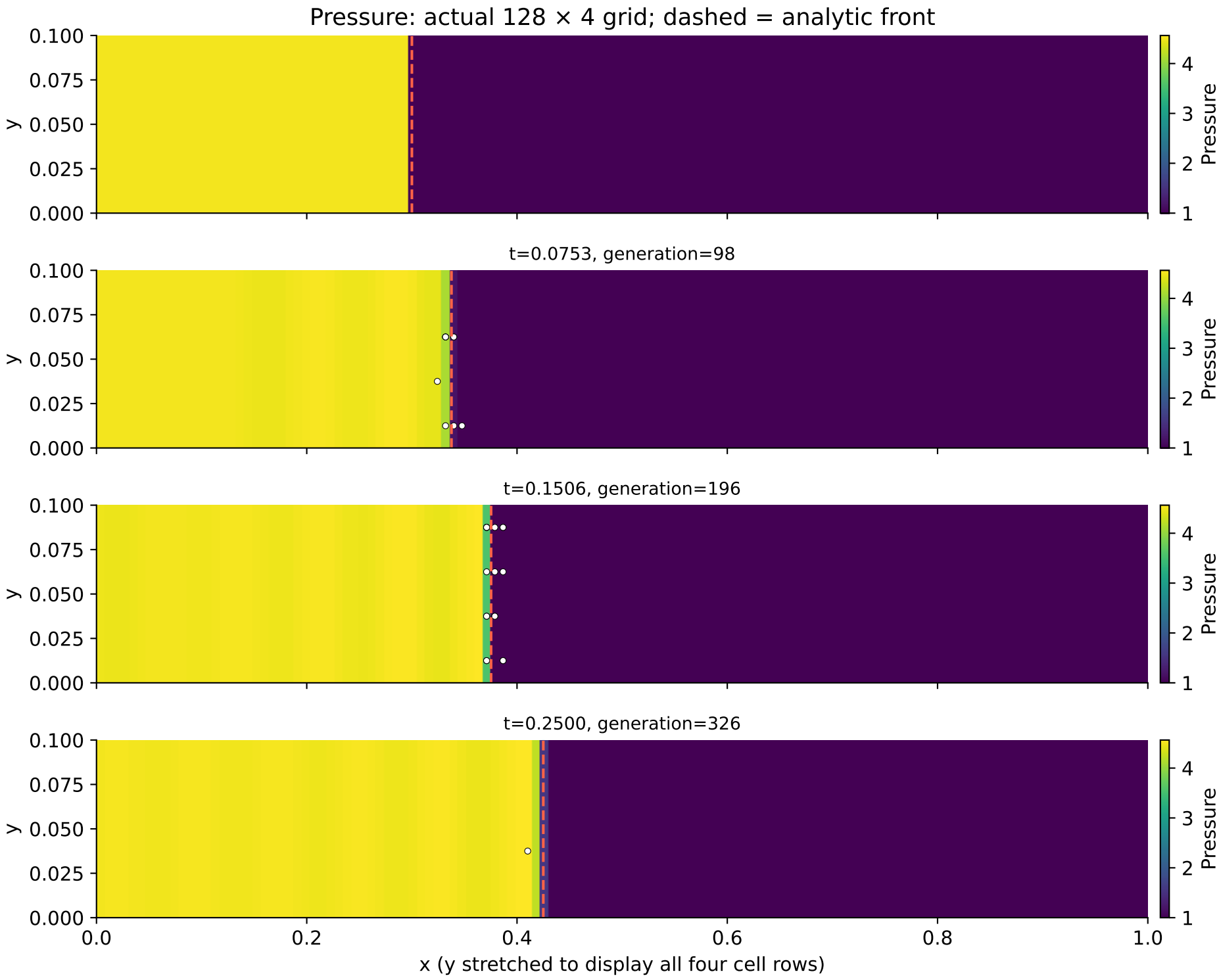
Read the plotted quantities with the notes below; code time and spatial code units apply.



Native 2D fields. White dots are sampled particles inside the unreduced shock region, not a reconstruction of the full mesh mask. Top panel is the initial state (t=0); later panels use the actual recorded generation/time. All four y rows are native grid values.

Figure 02 | 2d pressure

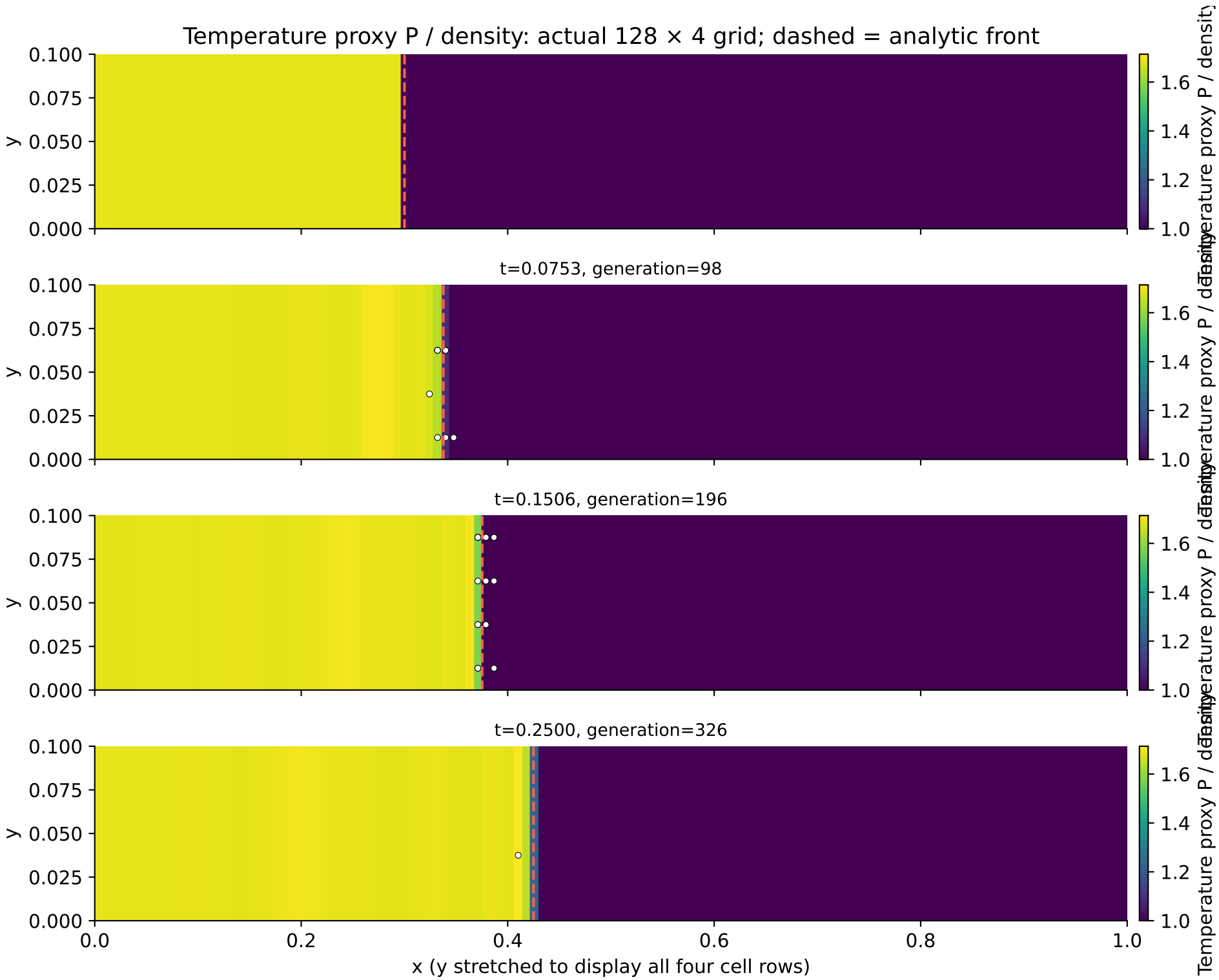
Read the plotted quantities with the notes below; code time and spatial code units apply.



Native 2D fields. White dots are sampled particles inside the unreduced shock region, not a reconstruction of the full mesh mask. Top panel is the initial state (t=0); later panels use the actual recorded generation/time. All four y rows are native grid values.

Figure 03 | 2d temperature

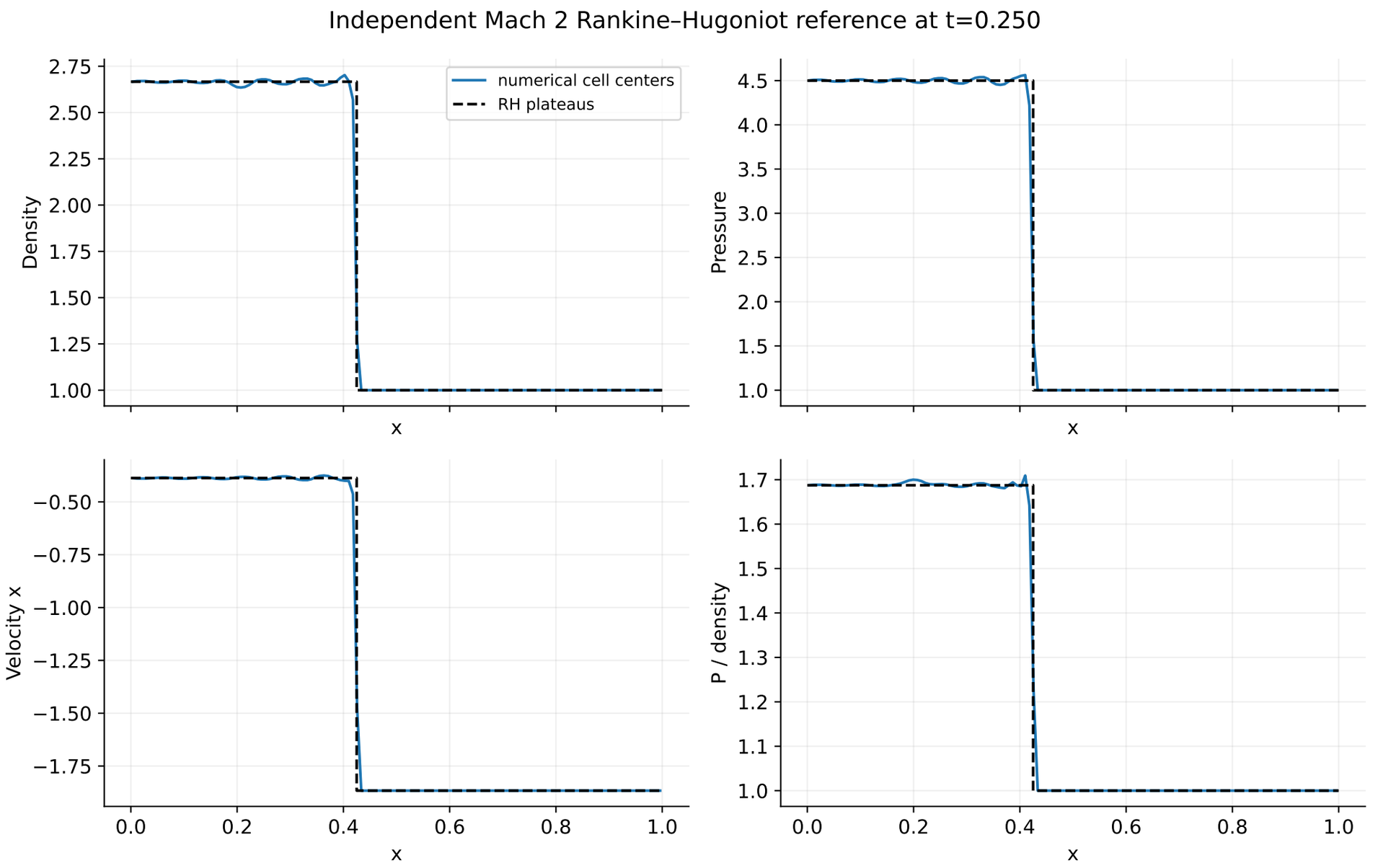
Read the plotted quantities with the notes below; code time and spatial code units apply.



Native 2D fields. White dots are sampled particles inside the unreduced shock region, not a reconstruction of the full mesh mask. Top panel is the initial state (t=0); later panels use the actual recorded generation/time. All four y rows are native grid values.

Figure 04 | rh profiles

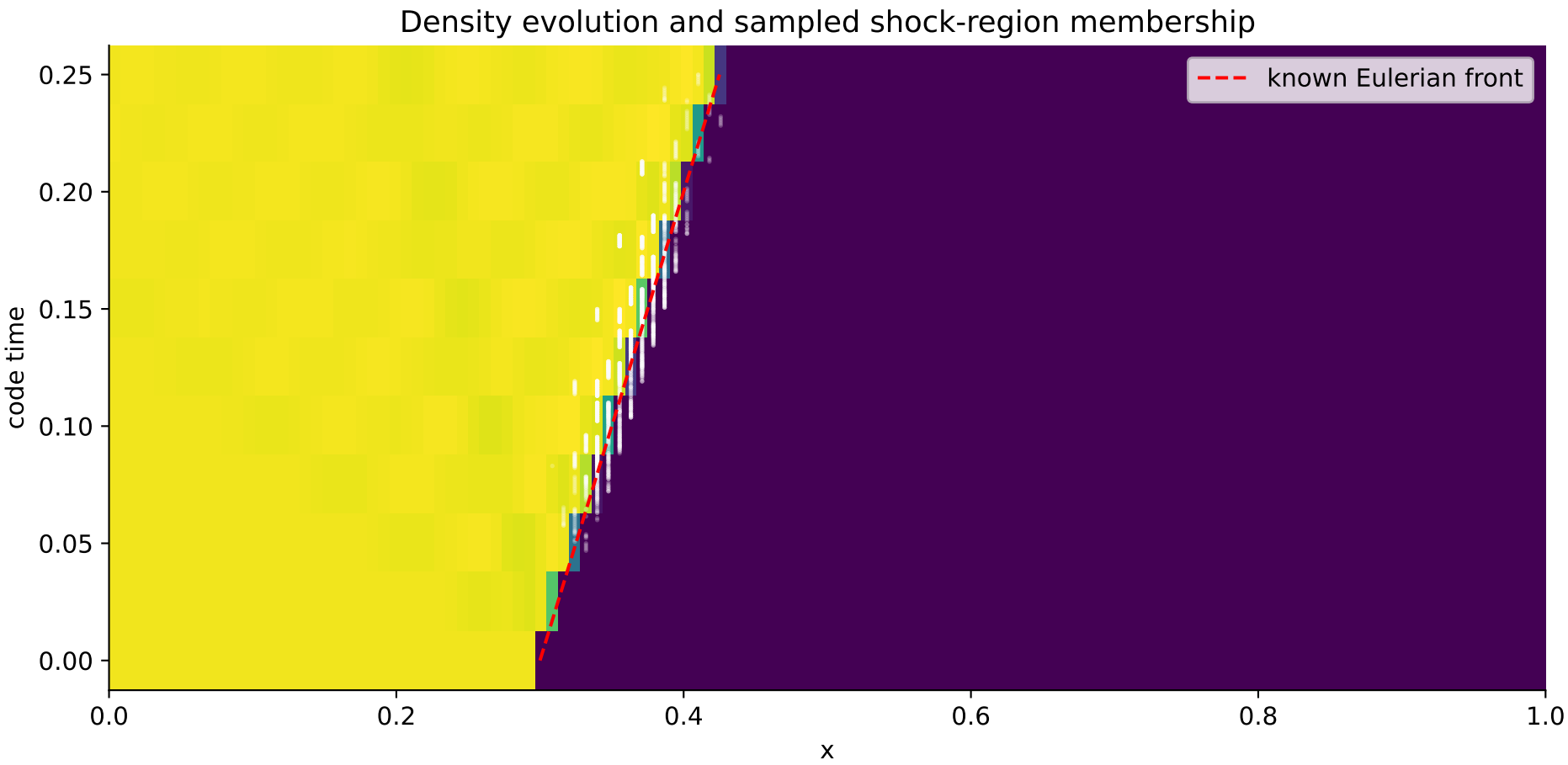
Read the plotted quantities with the notes below; code time and spatial code units apply.



Analytic discontinuity and plateaus versus resolved fluid profiles; finite shock thickness is visible. No strength acceptance tolerance is inferred.

Figure 05 | shock space time

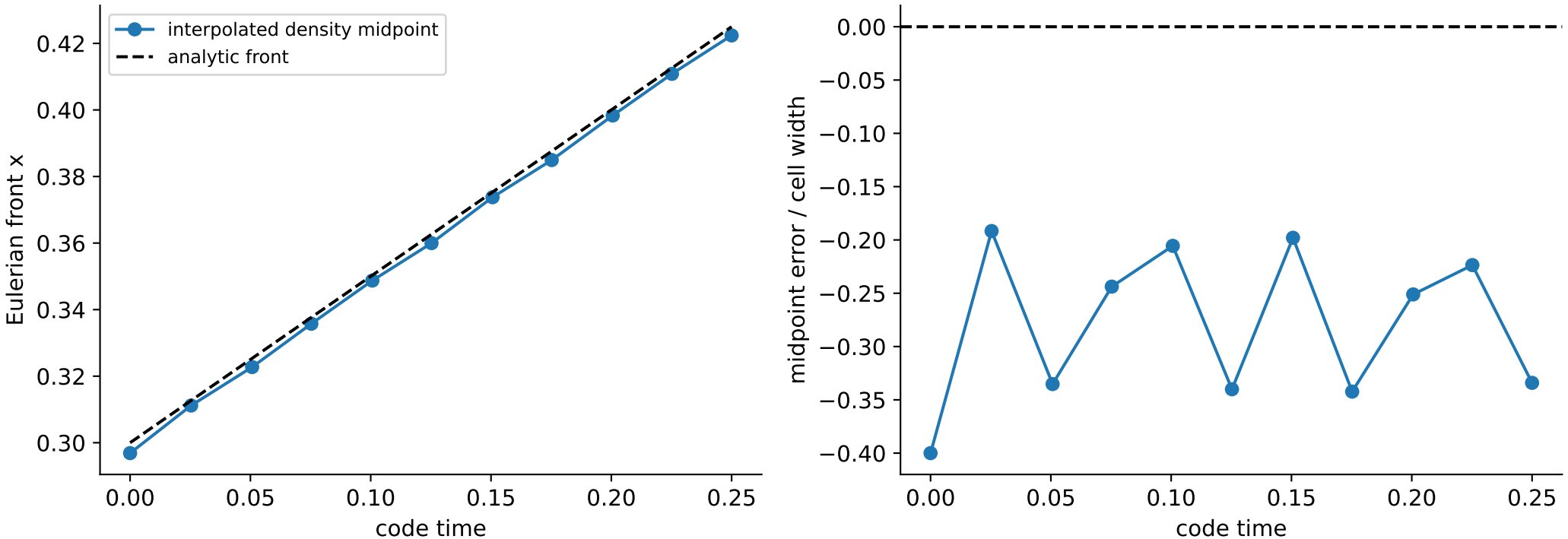
Read the plotted quantities with the notes below; code time and spatial code units apply.



White dots are synchronized sampled mask hits; the red line uses the prescribed shock speed, not a fitted particle crossing time.

Figure 06 | front position

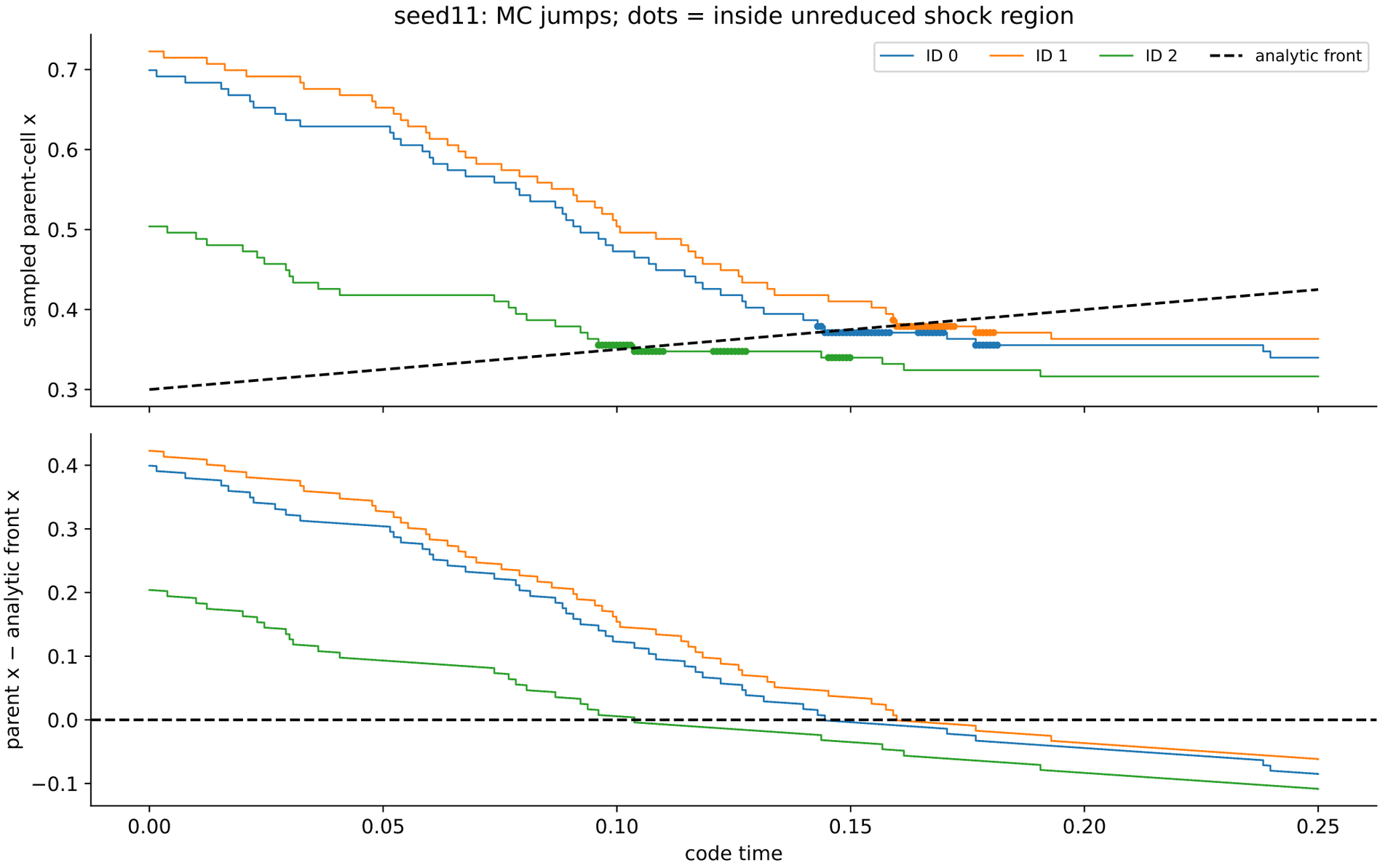
Read the plotted quantities with the notes below; code time and spatial code units apply.



Density-midpoint interpolation is an Eulerian diagnostic, not an encounter timing estimator. Initial cell placement contributes an offset.

Figure 07 | seed11 parents

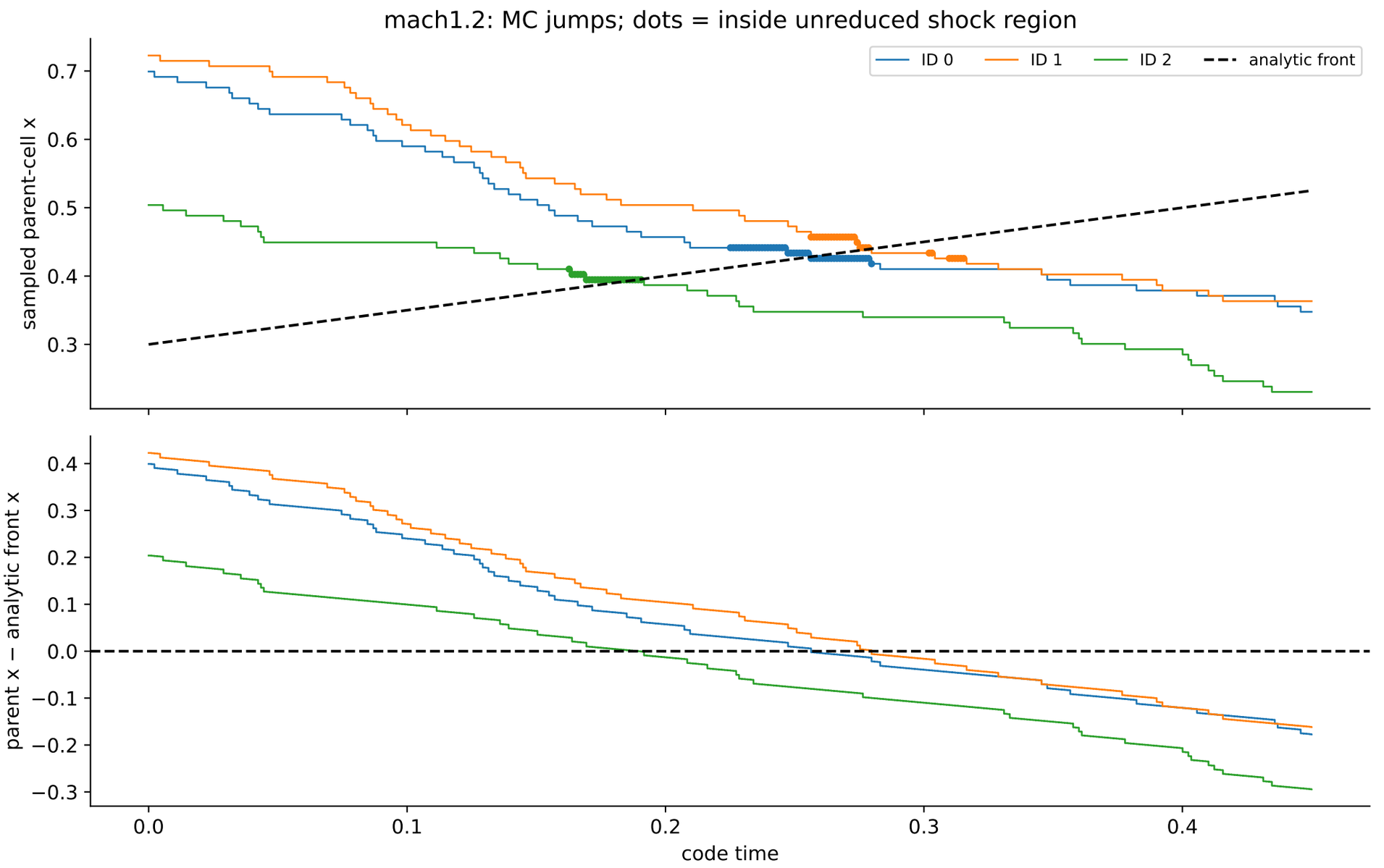
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 08 | mach1.2 parents

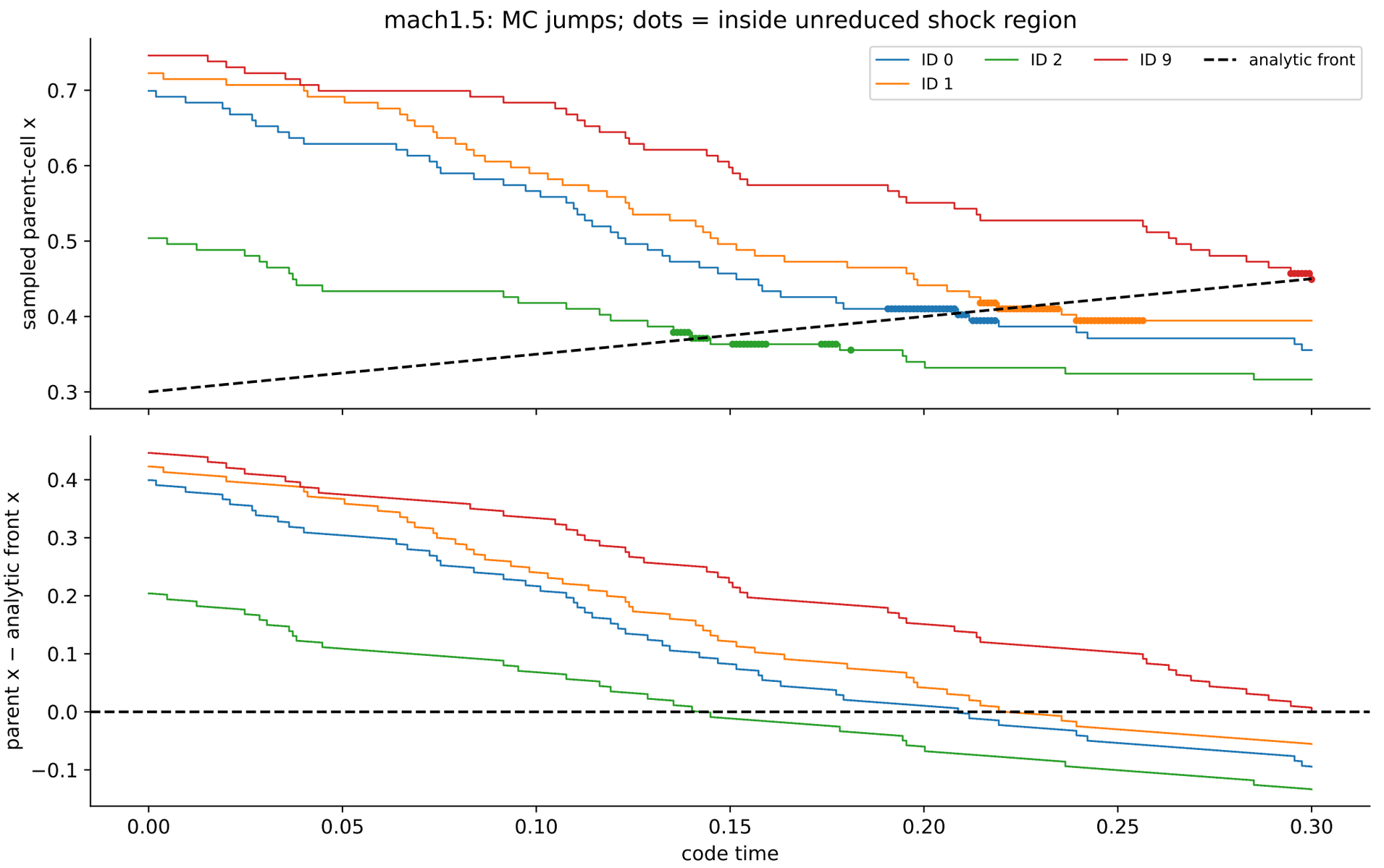
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=1.2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 09 | mach1.5 parents

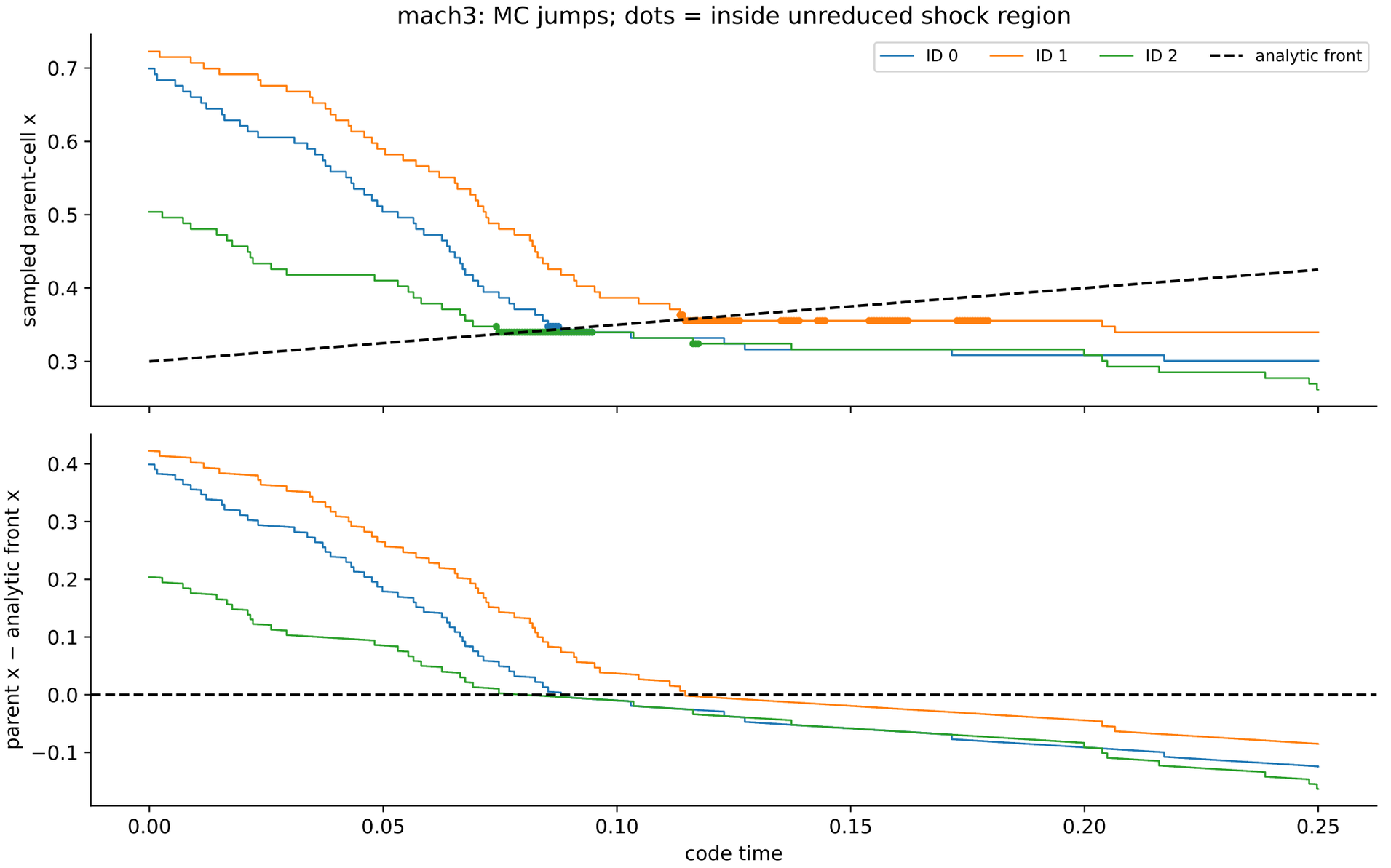
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=1.5; 127 complete episodes / 127 unique particles; 1 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 10 | mach3 parents

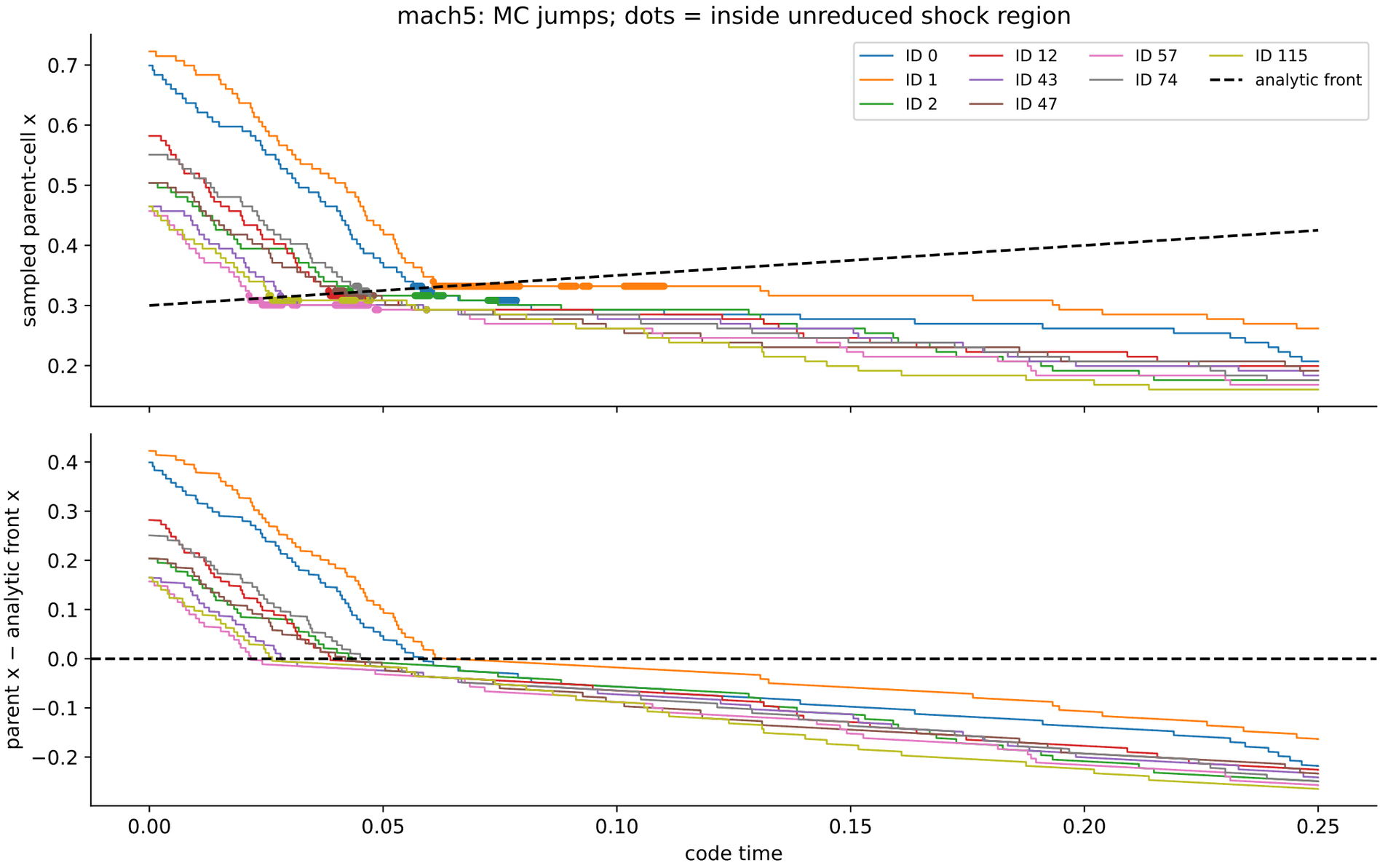
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=3.0; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 11 | mach5 parents

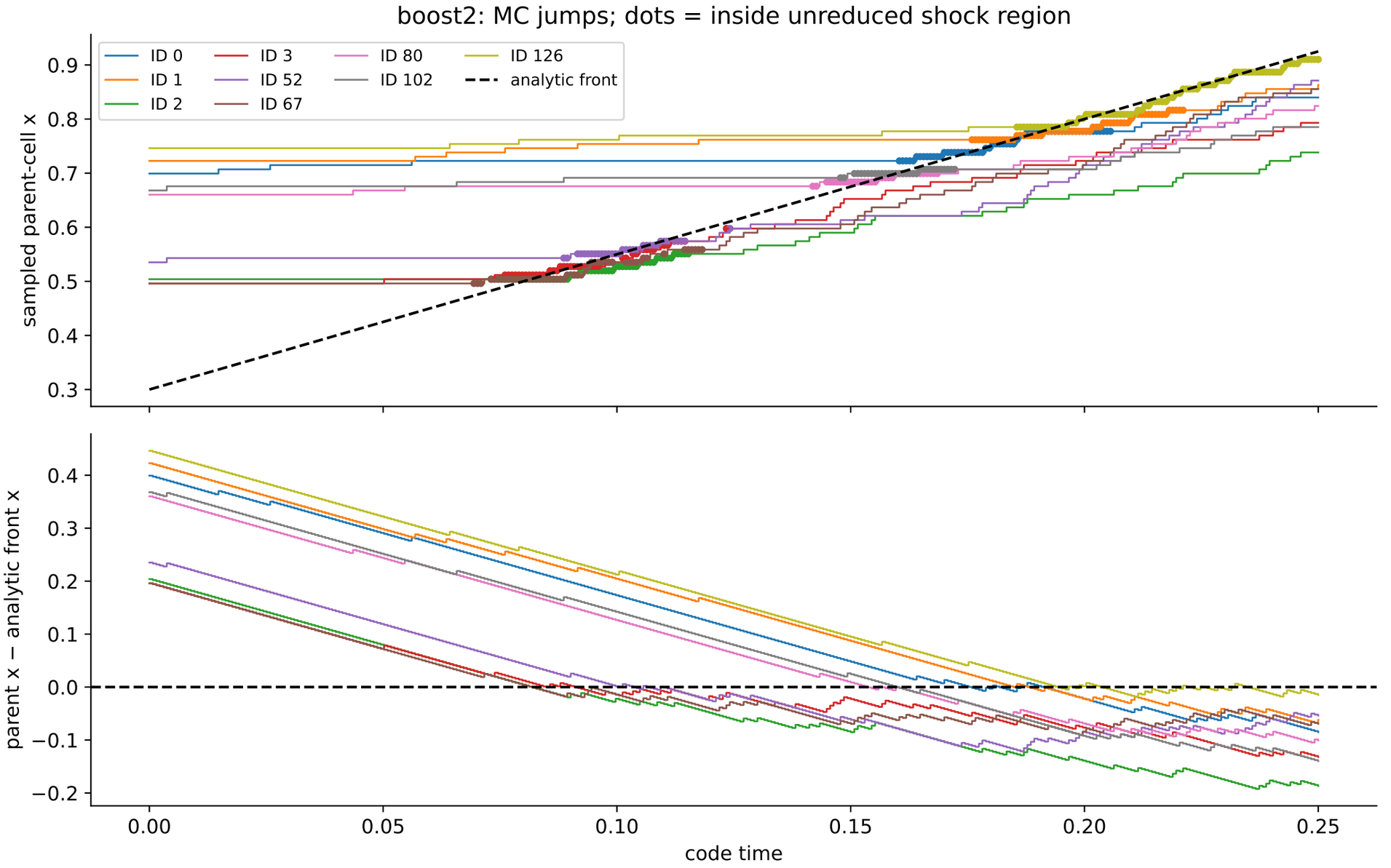
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=5.0; 134 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 12 | boost2 parents

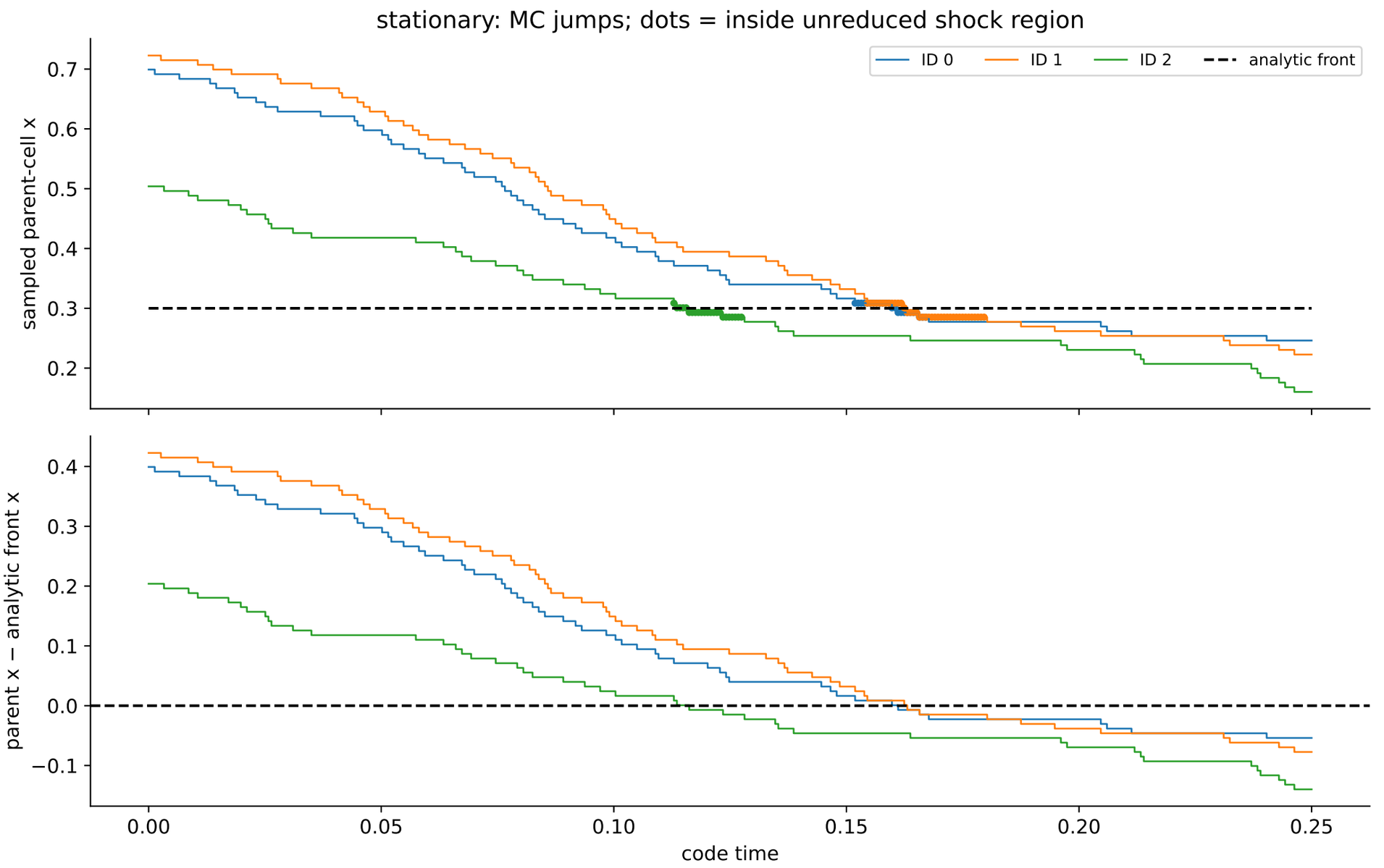
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=2; 132 complete episodes / 127 unique particles; 1 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 13 | stationary parents

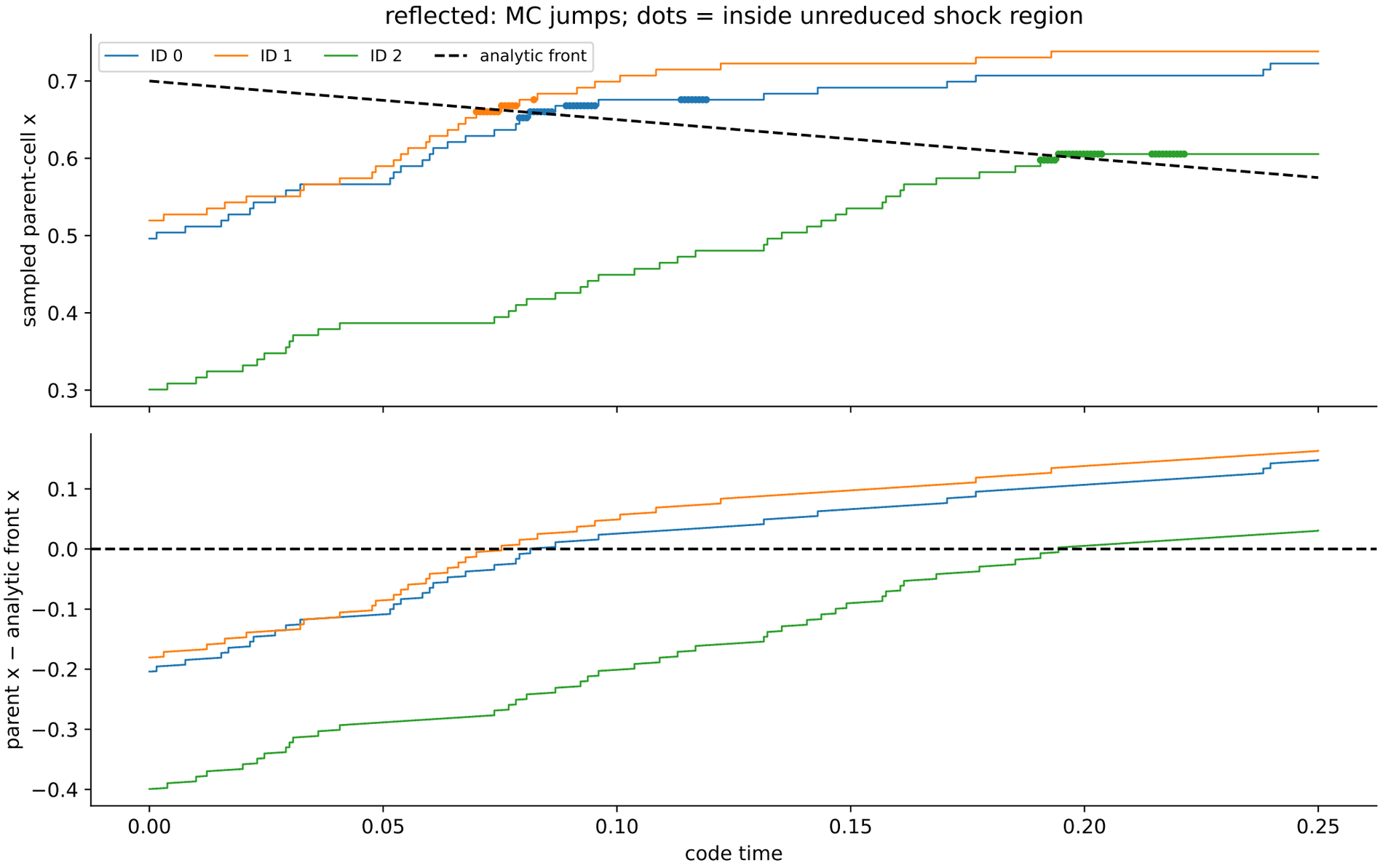
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 14 | reflected parents

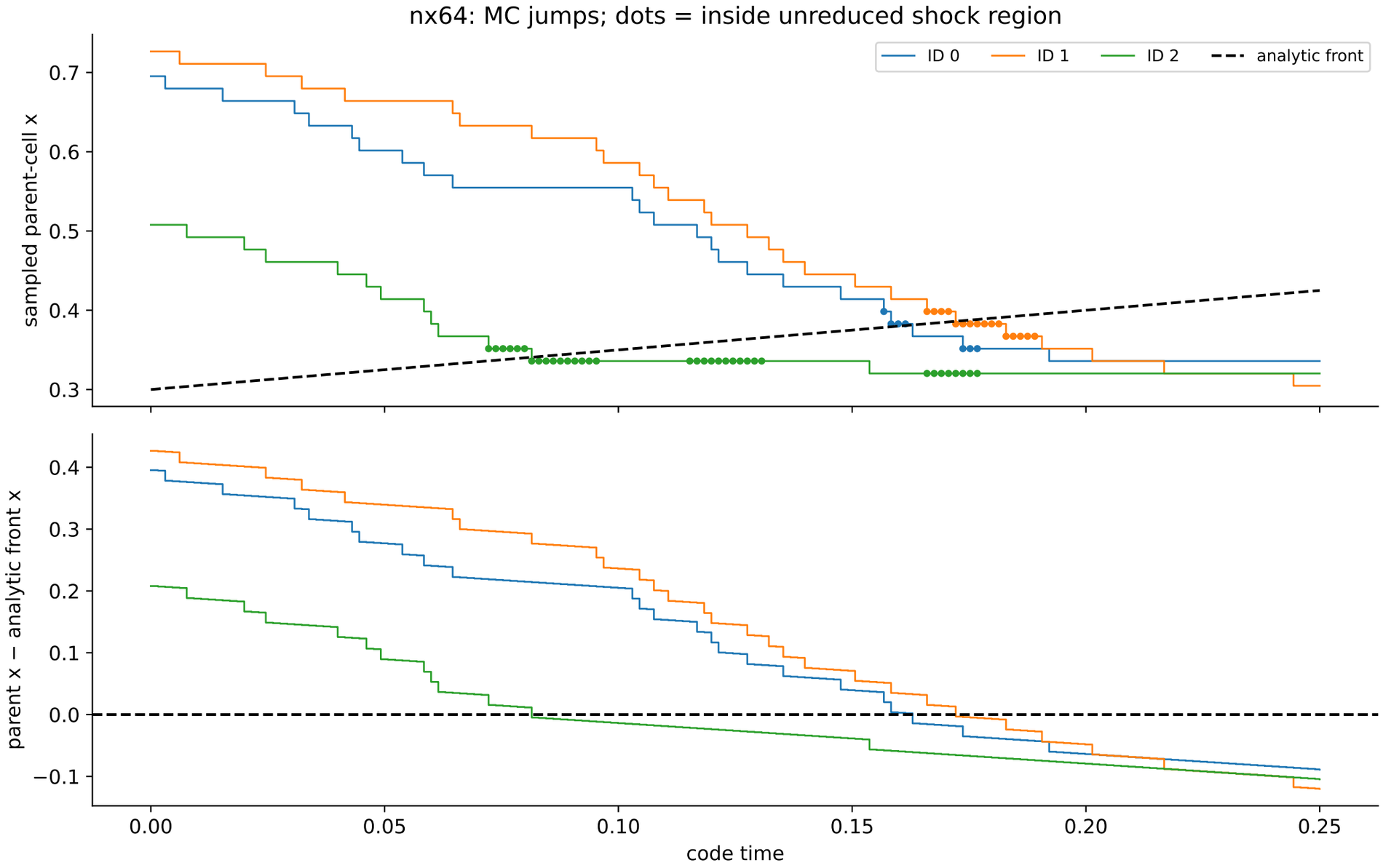
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 15 | nx64 parents

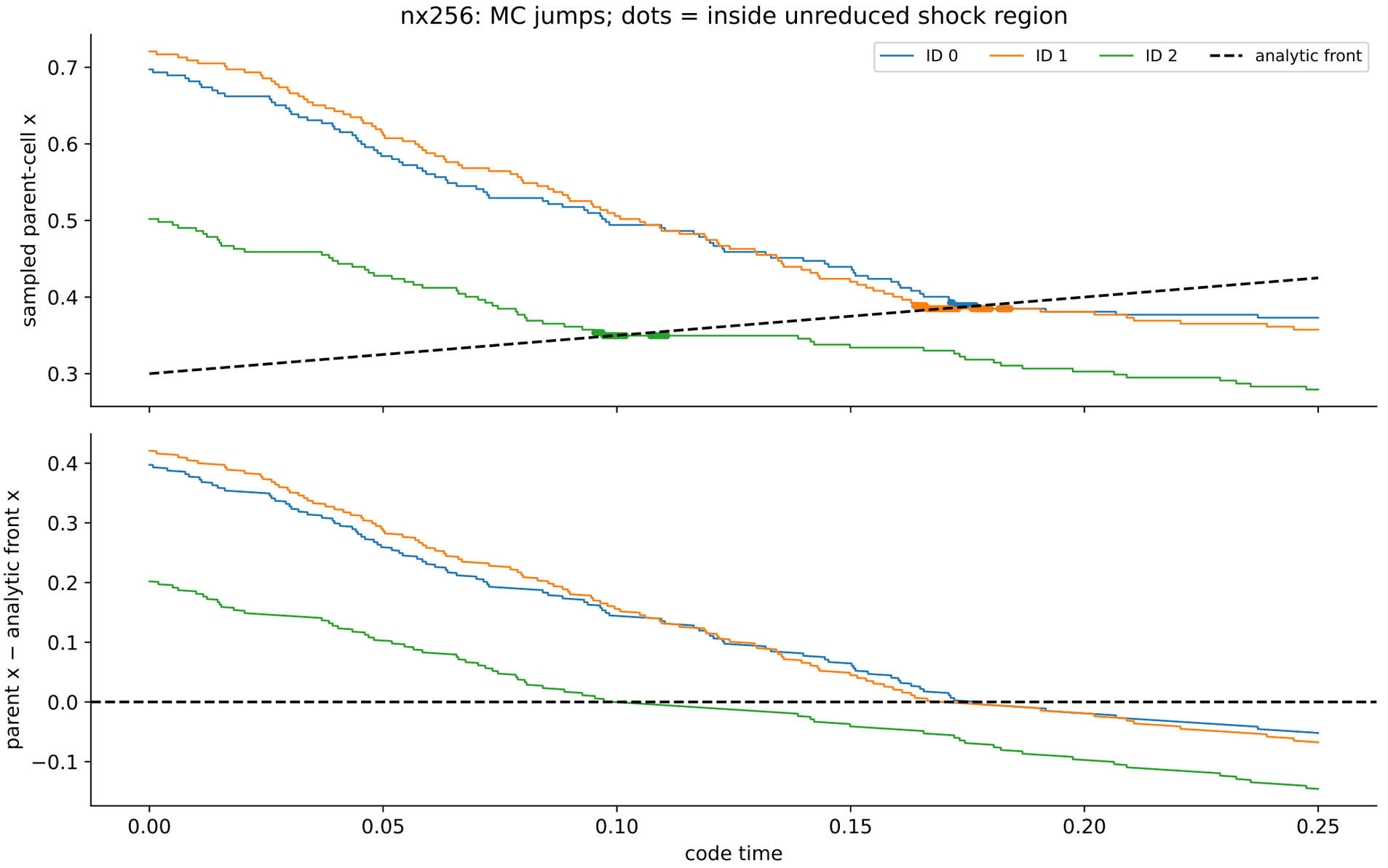
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 16 | nx256 parents

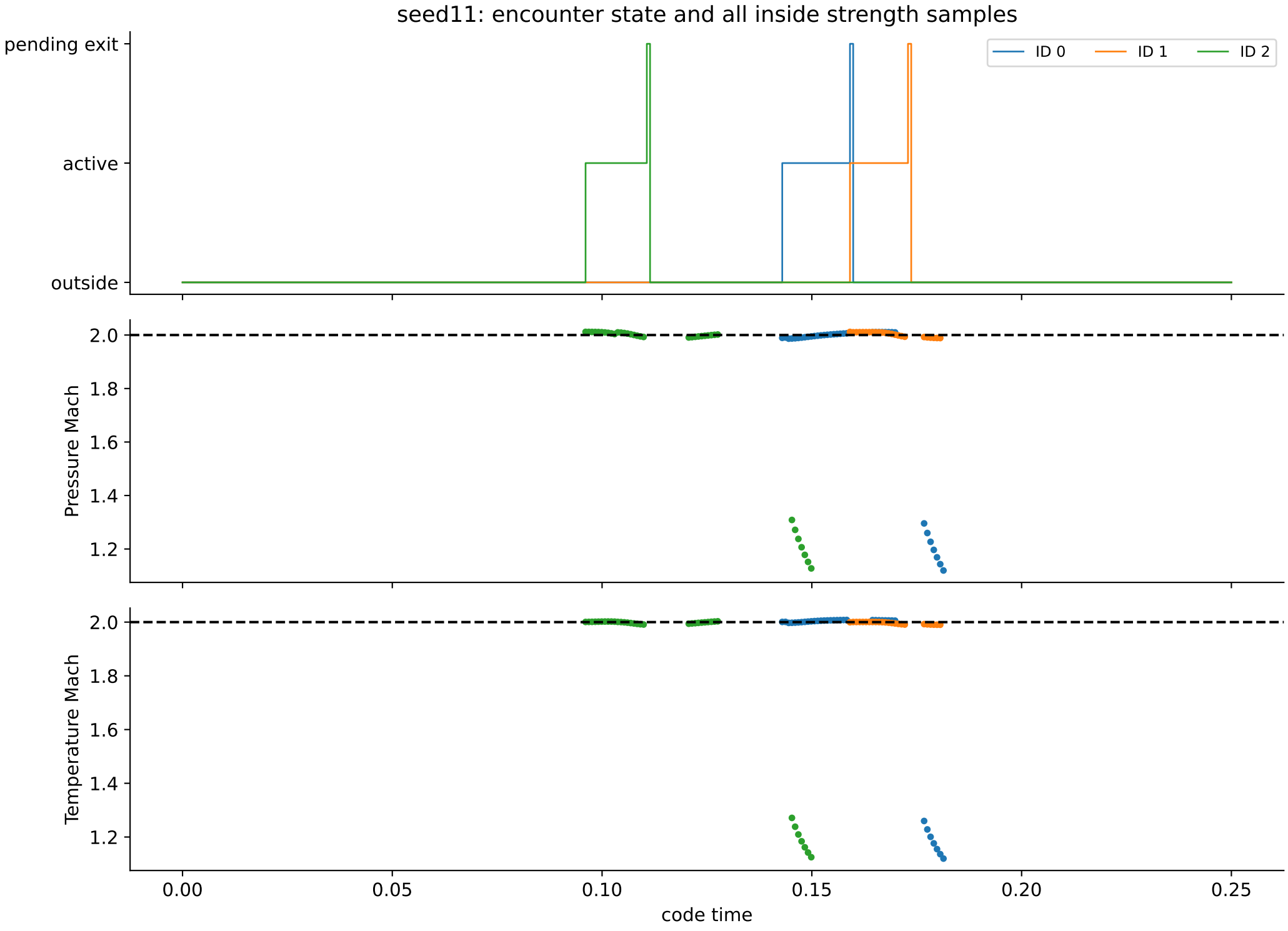
Read the plotted quantities with the notes below; code time and spatial code units apply.



MC positions are cell centers. Lines display discrete sample sequences, not continuous material trajectories or exact crossing times. Case: input Mach=2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 17 | seed11 strength time

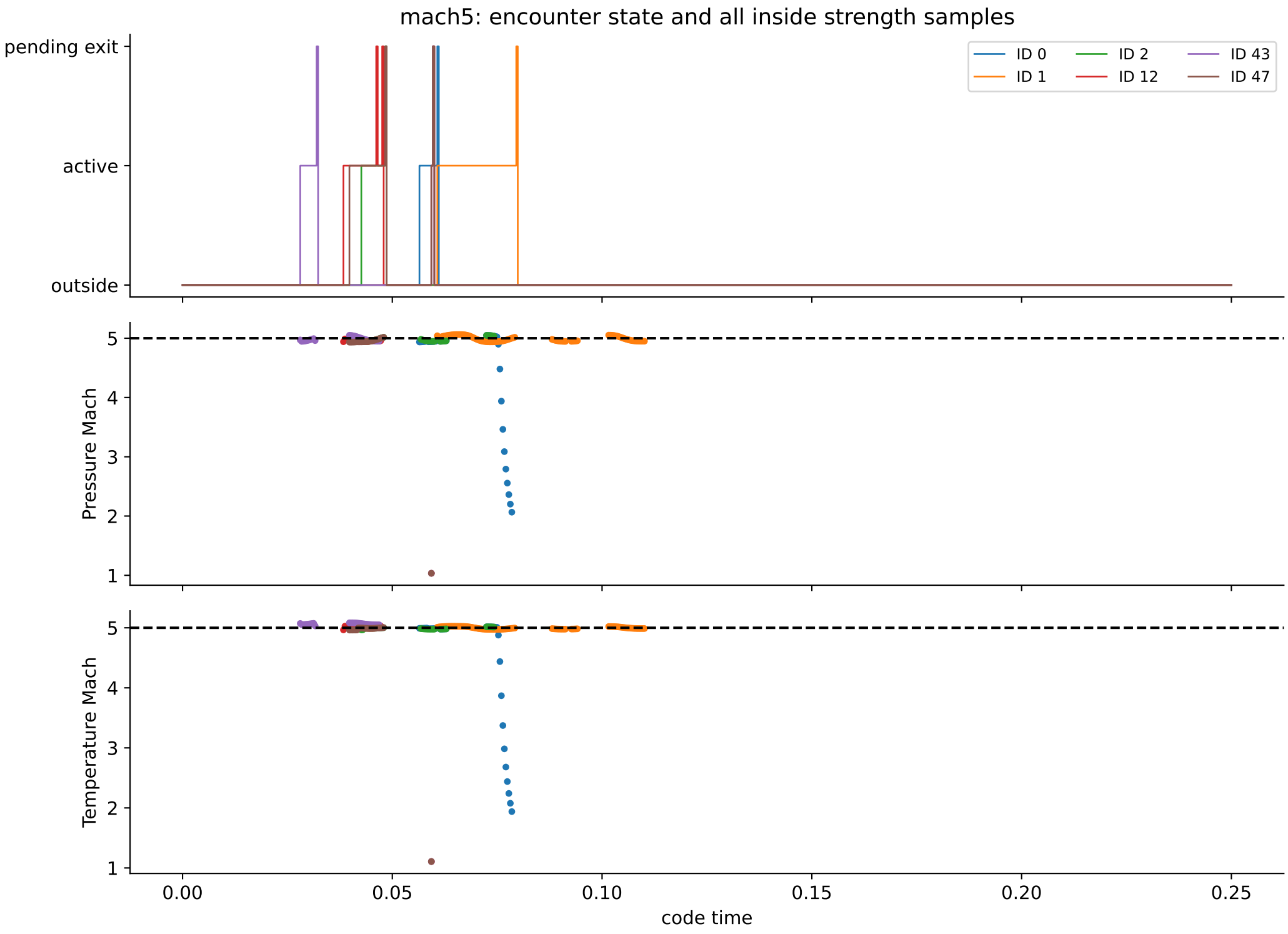
Read the plotted quantities with the notes below; code time and spatial code units apply.



Strength is based on fixed detector endpoints; these are not certified upstream/downstream plateaus. Gaps mean no inside sample. Case: input Mach=2; 128 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 18 | mach5 strength time

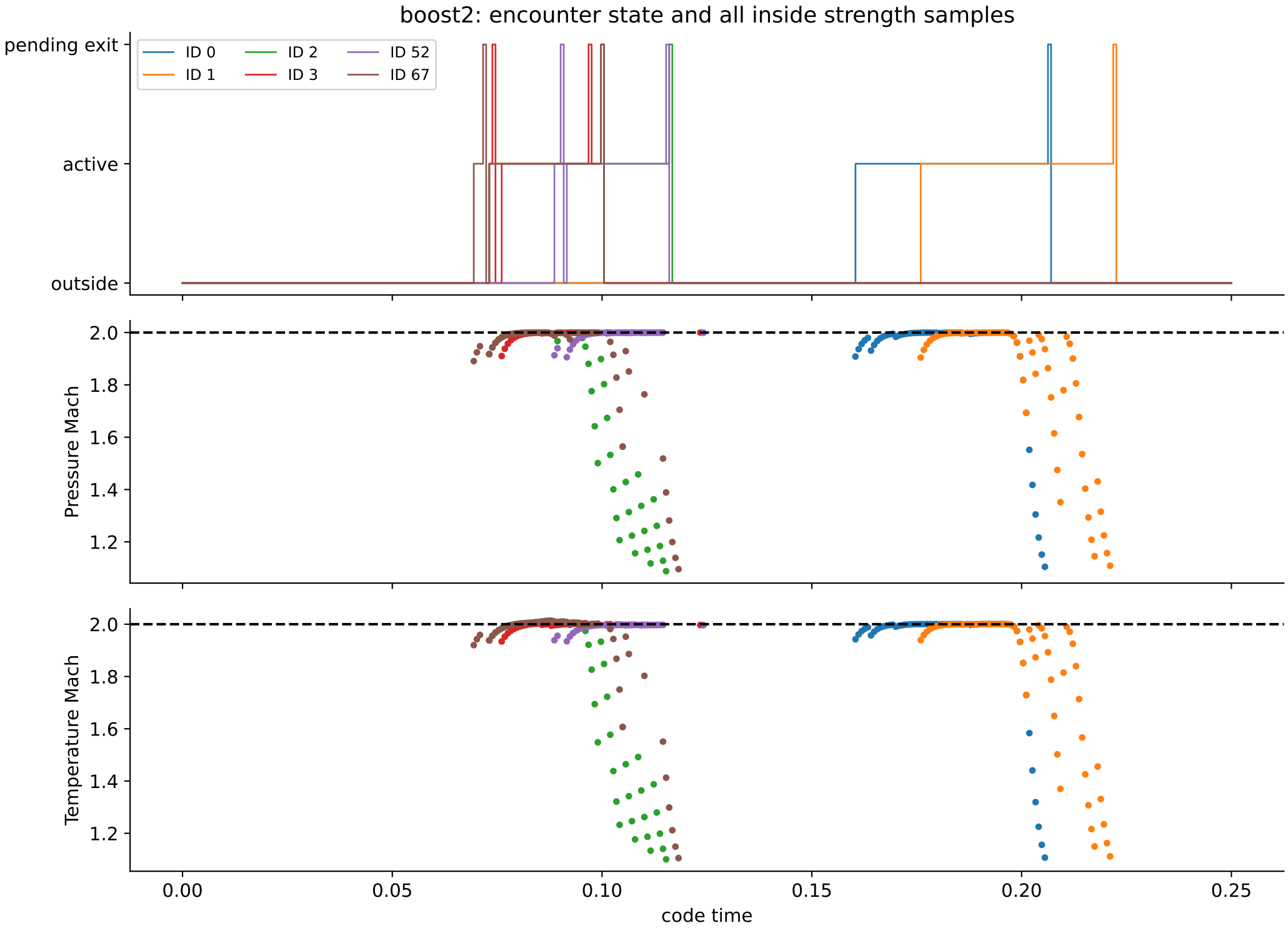
Read the plotted quantities with the notes below; code time and spatial code units apply.



Strength is based on fixed detector endpoints; these are not certified upstream/downstream plateaus. Gaps mean no inside sample. Case: input Mach=5.0; 134 complete episodes / 128 unique particles; 0 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 19 | boost2 strength time

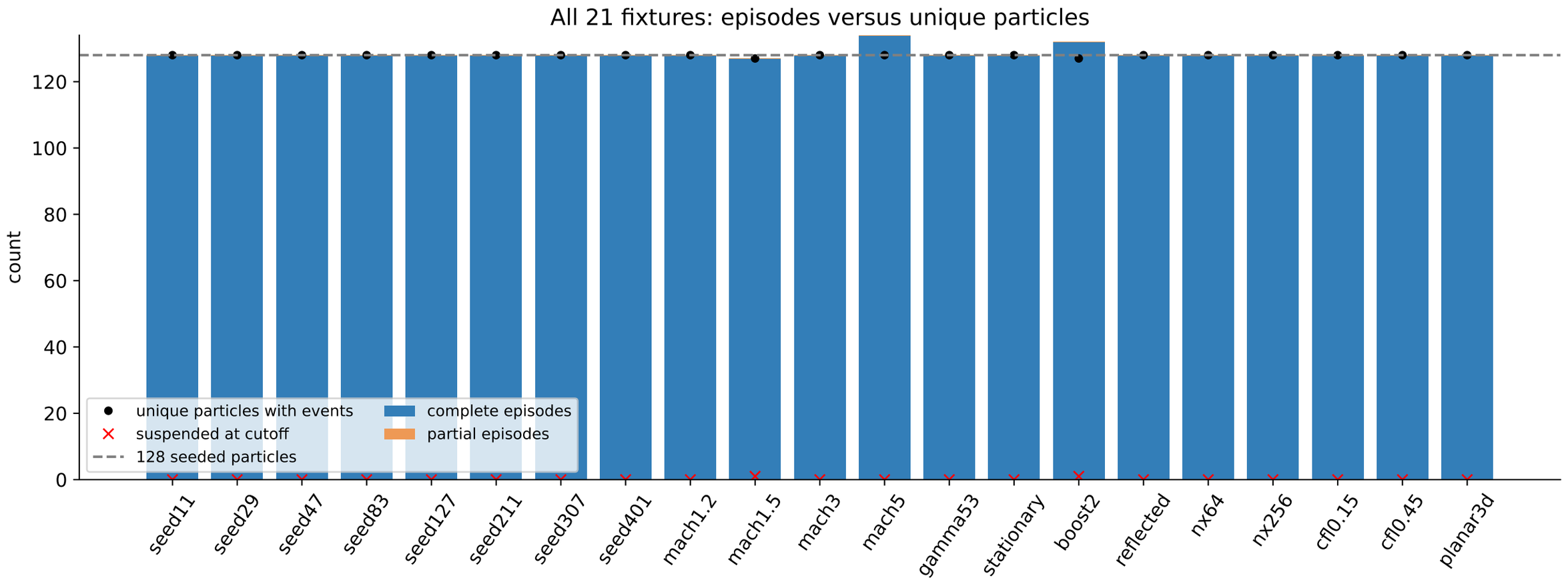
Read the plotted quantities with the notes below; code time and spatial code units apply.



Strength is based on fixed detector endpoints; these are not certified upstream/downstream plateaus. Gaps mean no inside sample. Case: input Mach=2; 132 complete episodes / 127 unique particles; 1 suspended. Each trace is a selected ID, not an ensemble-averaged material path.

Figure 20 | episode counts

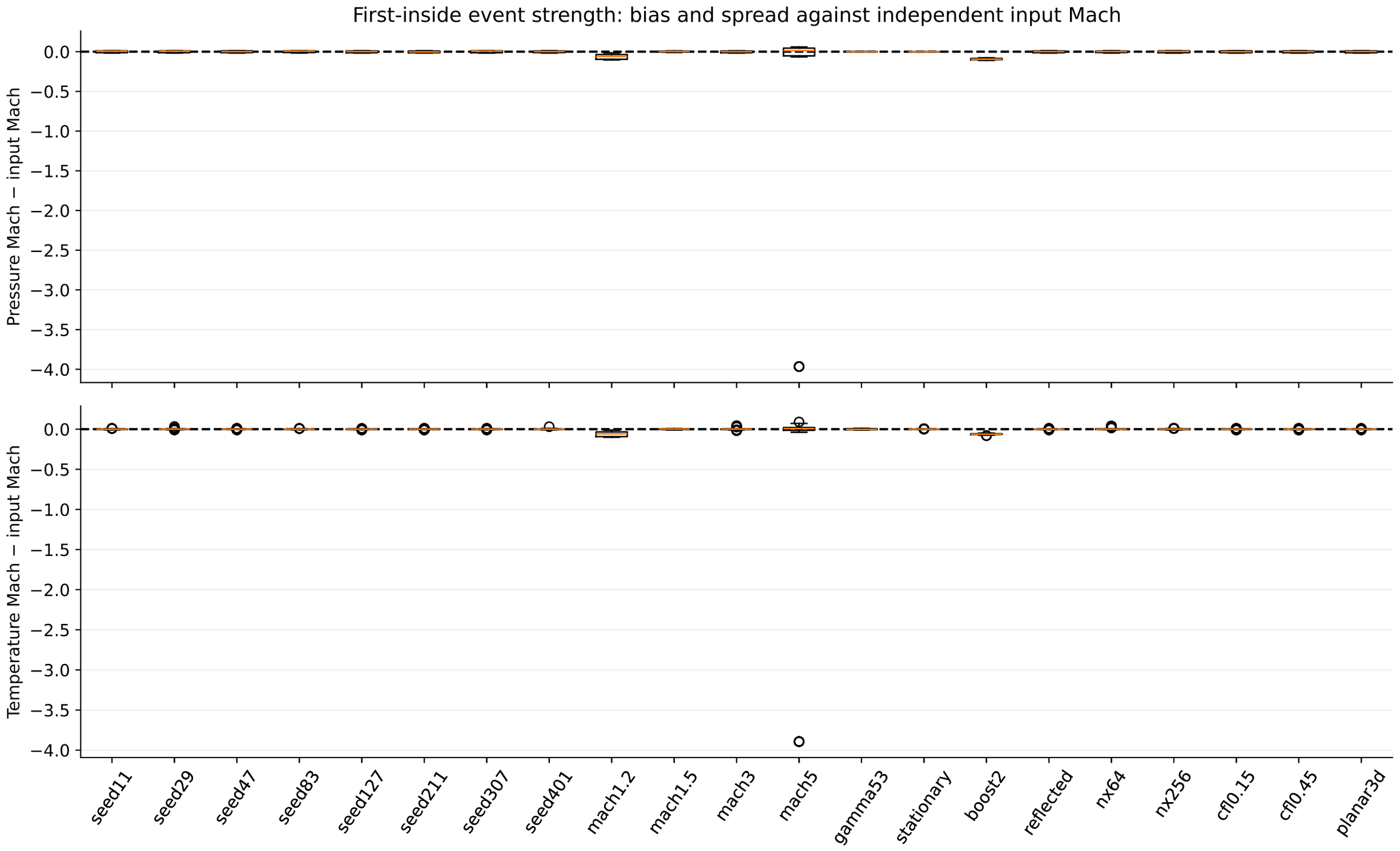
Read the plotted quantities with the notes below; code time and spatial code units apply.



Execution succeeded for all cases. Extra episodes and censored observations remain physical limitations, not scientific passes. All eight Mach 2 seeds recover 128/128; Mach 5 gives 134/128; boosted Mach 2 gives 132/127 plus one suspended.

Figure 21 | mach errors

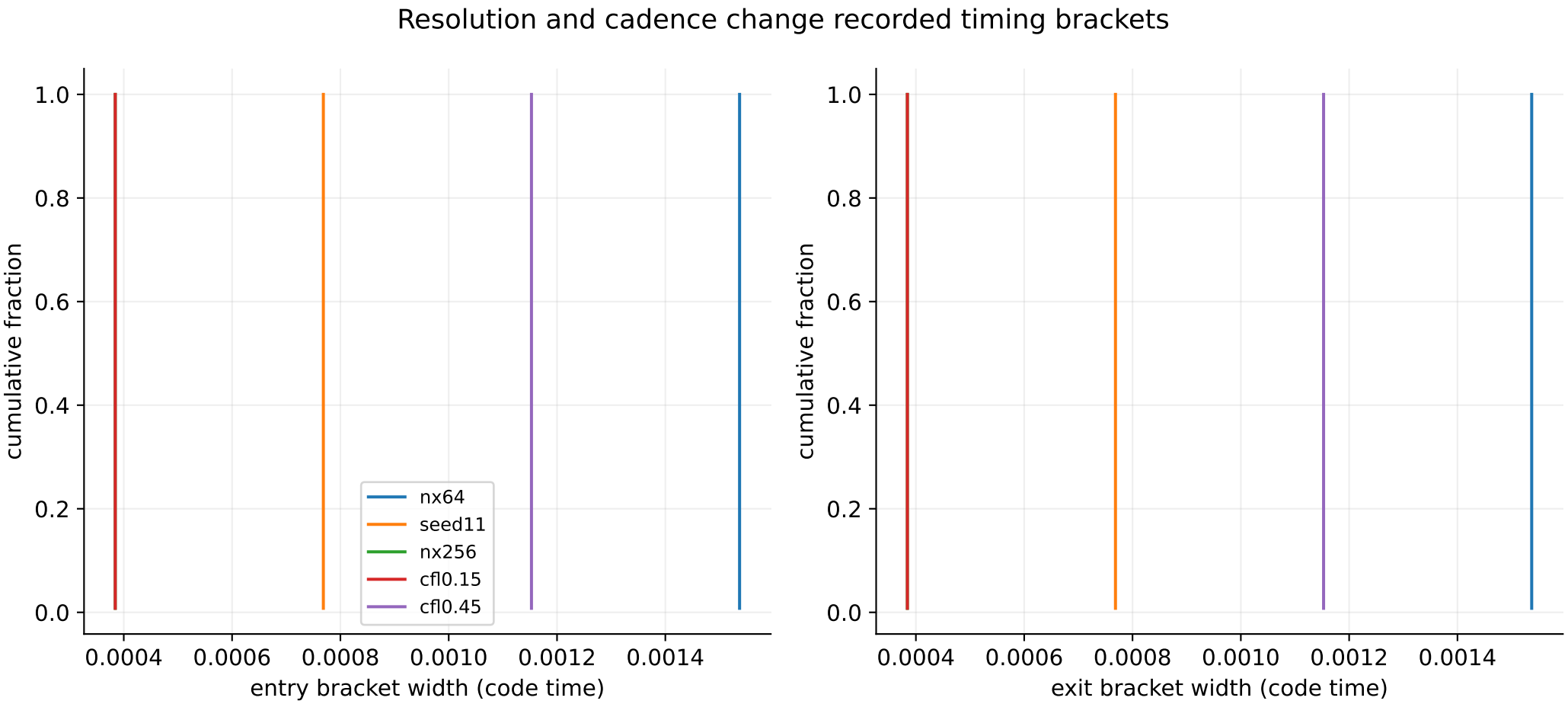
Read the plotted quantities with the notes below; code time and spatial code units apply.



Every emitted event contributes, including weak secondary lobes. No outlier removal, calibrated endpoint selection or relaxed tolerance is applied.

Figure 22 | timing bracket widths

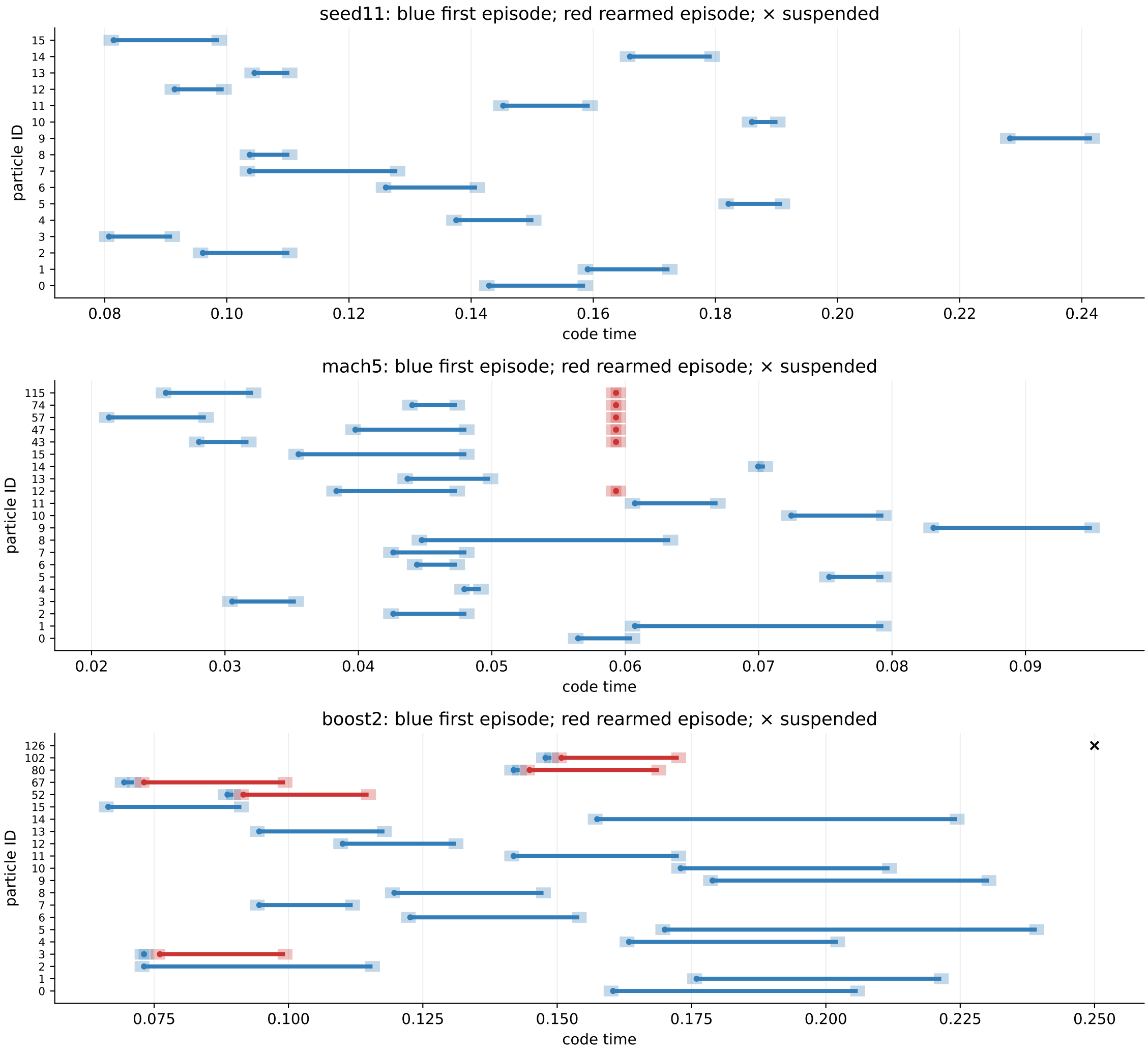
Read the plotted quantities with the notes below; code time and spatial code units apply.



Brackets bound adjacent region-membership observations. Their widths are not errors against an invented exact MC crossing time. Coincident curves can overlap: halving CFL and doubling x resolution give similar observation cadences in these fixtures.

Figure 23 | encounter brackets

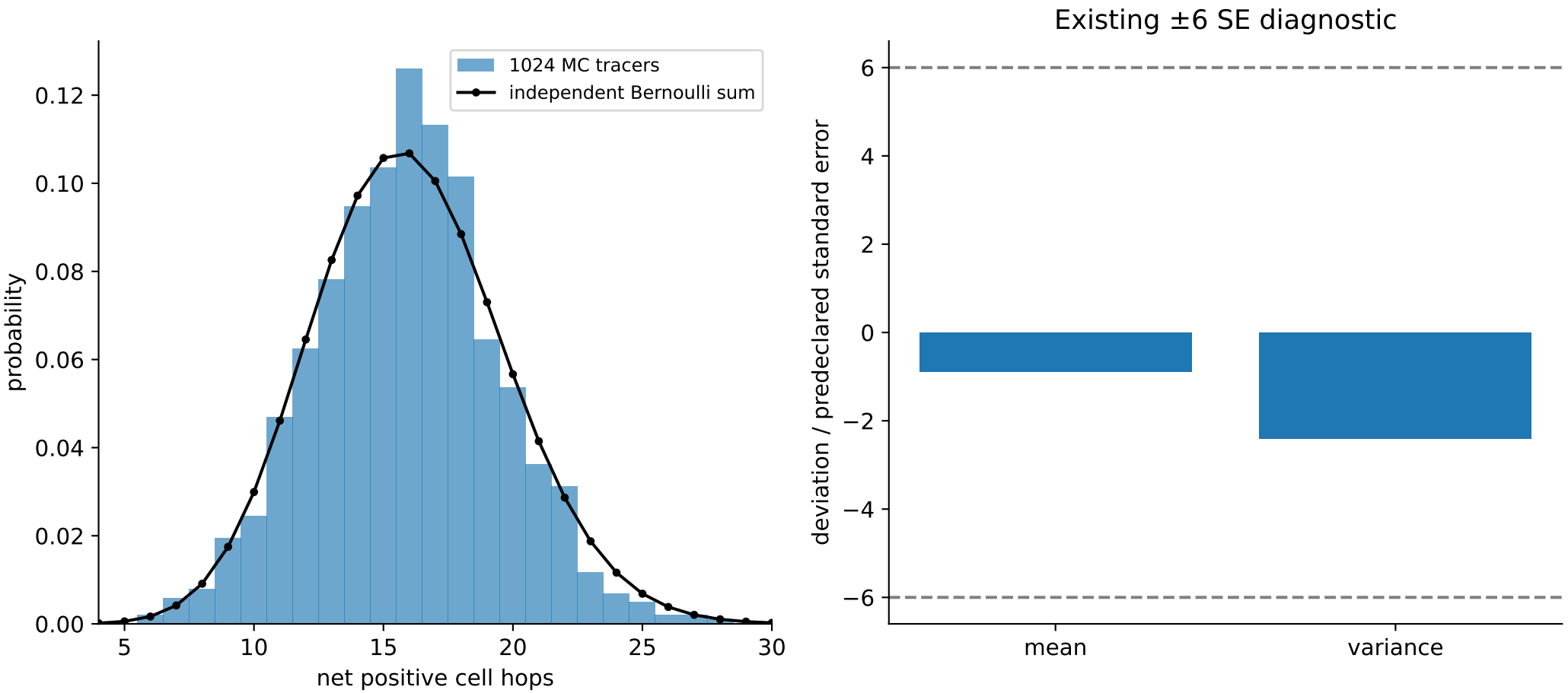
Read the plotted quantities with the notes below; code time and spatial code units apply.



Thick pale ends show recorded entry/exit brackets; narrow solid segments join first/last inside observations, not exact physical residence.

Figure 24 | uniform transport

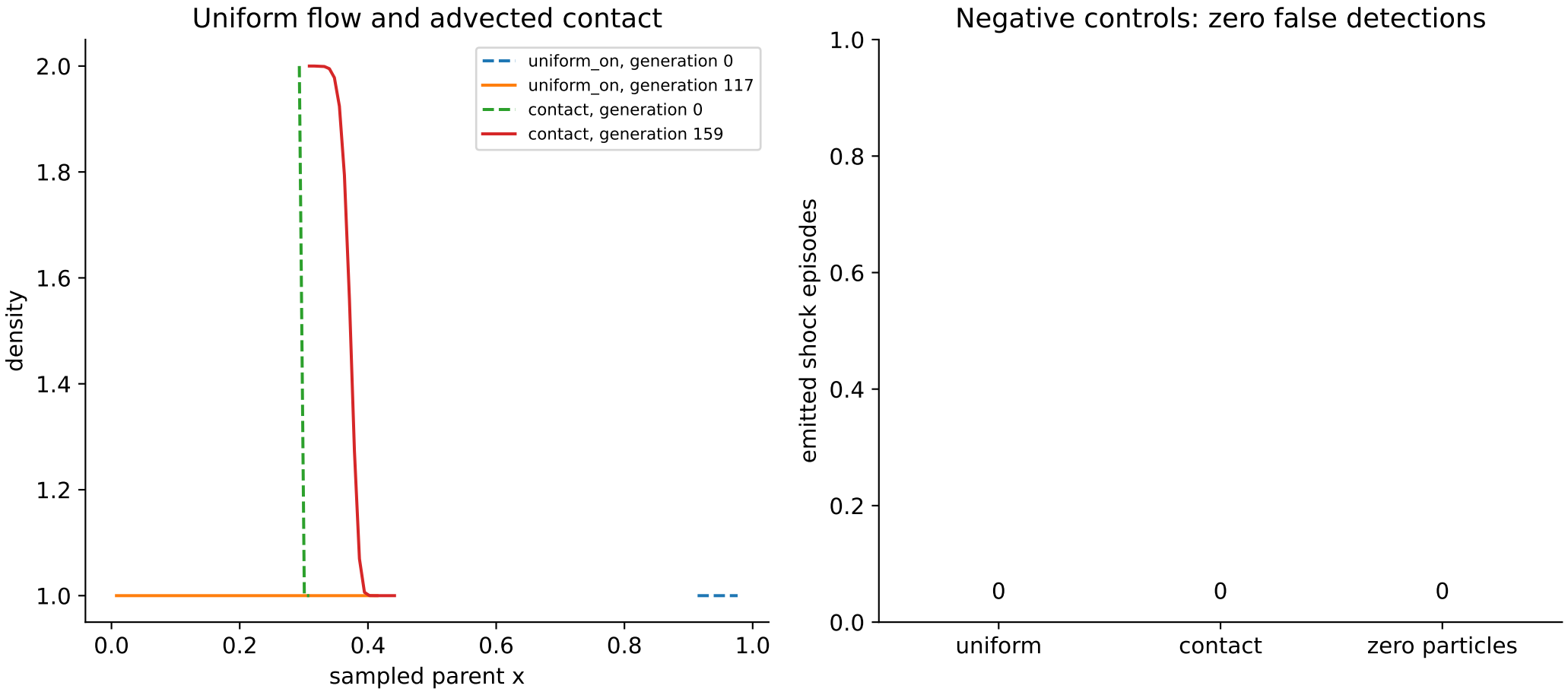
Read the plotted quantities with the notes below; code time and spatial code units apply.



Physical flux probabilities define this independent ensemble reference. A fixed seed diagnostic does not prove RNG independence at research scales. The reference is a sum of per-step Bernoulli probabilities. The measured mean/variance deviations are -0.89/-2.40 standard errors.

Figure 25 | negative controls

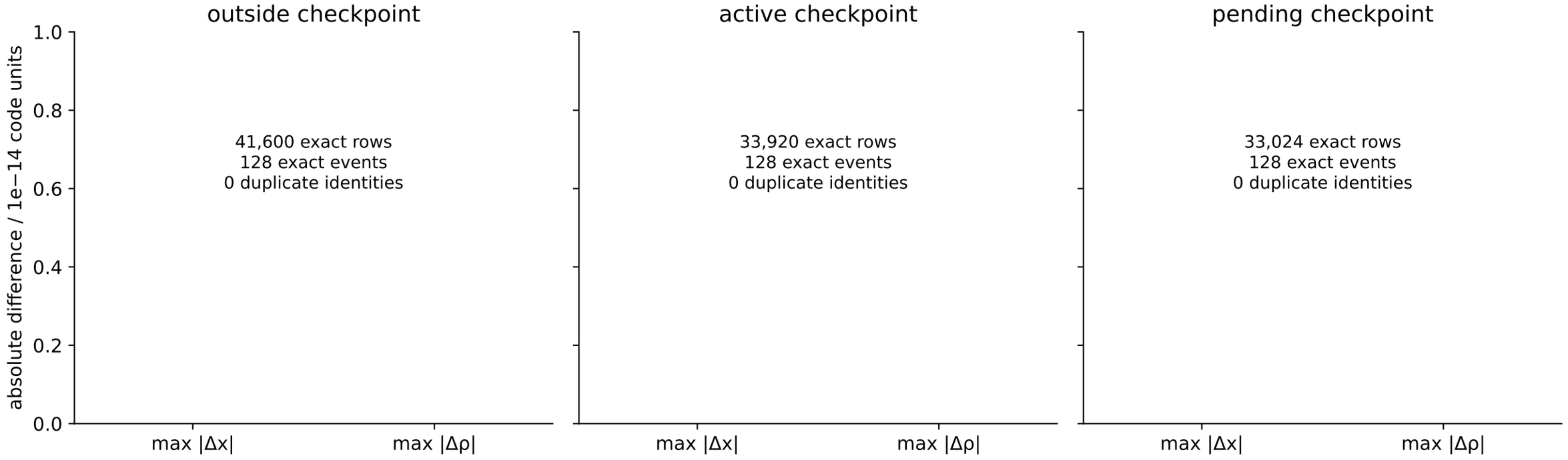
Read the plotted quantities with the notes below; code time and spatial code units apply.



Contact samples show a density change at constant pressure and velocity. Uniform/contact/empty fixtures emitted no events.

Figure 26 | restart agreement

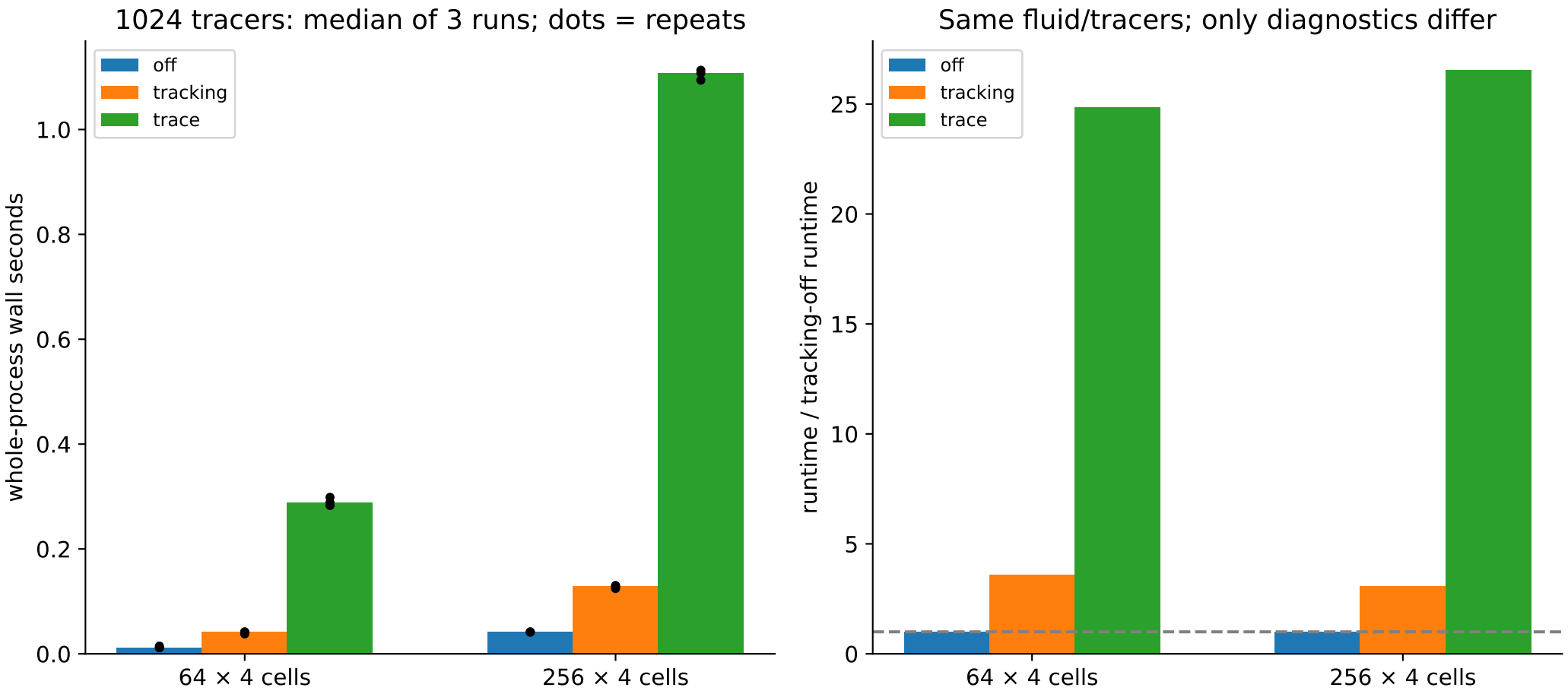
Read the plotted quantities with the notes below; code time and spatial code units apply.



Replay from outside, active and pending-exit states preserved exact same-partition particle rows and event identities. No cross-rank claim. All plotted differences are exactly zero. The 1e-14 display unit is not a tolerance; comparisons use exact equality.

Figure 27 | runtime cost

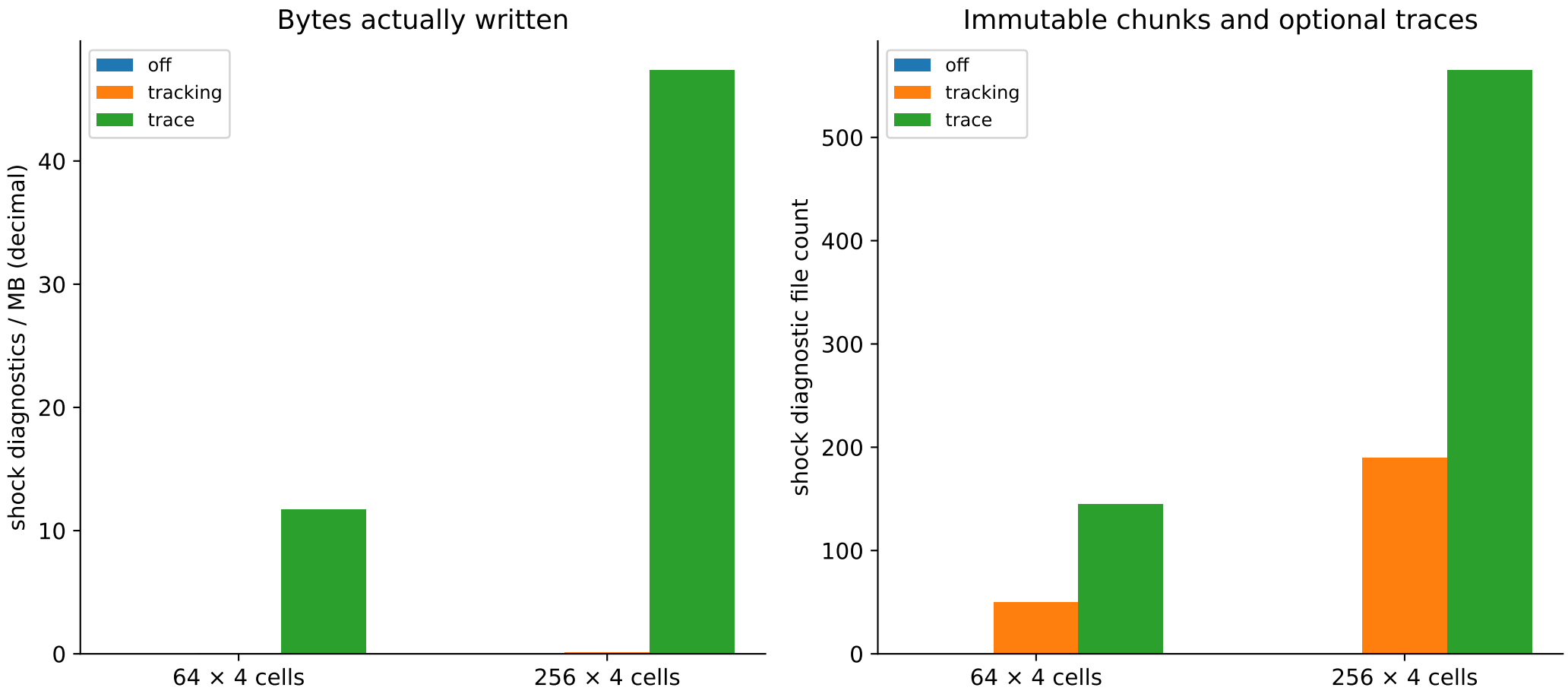
Read the plotted quantities with the notes below; code time and spatial code units apply.



Release Serial CPU, uniform flow, tlim=0.1. Startup, durable empty event chunks and output are included. These are laptop costs, not cluster budgets.

Figure 28 | output cost

Read the plotted quantities with the notes below; code time and spatial code units apply.



Actual directory sizes/counts exclude fluid and thermo output. Trace mode records every observation/transfer and is intended for small qualification jobs.